

# boba\_extension\_dim10

## Dimension 10: Extension UI/UX Design Patterns - Comprehensive Research

**Date:** 2025-07-28 **Scope:** Browser extension UI/UX patterns for Boba torrent helper extension (Manifest V3) **Target:** Chrome/Edge (Chromium-based browsers)

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## 1. Popup Page Design

### 1.1 Design Constraints & Best Practices

Claim: Chrome enforces hard limits on popup dimensions: 800x600px maximum, 25x25px minimum. The sweet spot for most extensions is 380-450px wide.

Source: Extension Booster / Chromium Source Code

URL: <https://extensionbooster.net/blog/chrome-extension-popup-ui-design-best-practices-guide/>

Date: 2026-04-23

Excerpt: "Chrome enforces hard limits on popup dimensions: 800x600px maximum, 25x25px minimum. But those limits aren't design targets -- they're walls... The sweet spot that top extensions actually use is 400x500px."

Context: Documented in Chromium source as ``static constexpr gfx::Size kMaxSize = {800, 600};``

Confidence: high

Claim: Popup should use CSS reset, explicit body width (not 100%), and `overflow-y: auto` for scrolling. System font stack is recommended.

Source: Reintech Chrome Extension UI Guide

URL: <https://reintech.io/blog/building-user-interfaces-chrome-extensions>

Date: 2026-03-24

Excerpt: "A typical popup should be 300-400px wide and 400-600px tall. Going wider can feel awkward, and taller popups may get cut off on smaller screens."

Context: Use system-ui font family for native feel across platforms

Confidence: high

## 1.2 Popup HTML Structure

```
<!-- popup/popup.html -->
<!DOCTYPE html>
<html lang="en" data-theme="auto">
<head>
  <meta charset="UTF-8" />
  <meta name="color-scheme" content="light dark" />
  <meta name="viewport" content="width=device-width, initial-
    scale=1.0" />
  <title>Boba Torrent Helper</title>
  <link rel="stylesheet" href="popup.css" />
</head>
<body>
  <div id="app">
    <!-- Header: Fixed, non-scrolling -->
    <header class="popup-header">
      <div class="header-left">
        
        <span class="header-title" data-i18n="popupTitle">Boba</
          span>
      </div>
      <div class="header-actions">
        <button id="refresh-btn" class="icon-btn" aria-
          label="Refresh torrents" title="Scan page">
          <svg width="16" height="16" viewBox="0 0 16 16"
            fill="none" aria-hidden="true">
```

```

        <path d="M13.5 8A5.5 5.5 0 0 1 3.3 10.5M2.5 8a5.5 5.5
0 0 1 10.2-2.5M11.5 2.5V5h-2.5M4.5 13.5V11h2.5"
stroke="currentColor" stroke-width="1.5" stroke-
linecap="round" stroke-linejoin="round"/>
    </svg>
</button>
<button id="settings-btn" class="icon-btn" aria-
label="Open settings" title="Settings">
    <svg width="16" height="16" viewBox="0 0 16 16"
fill="none" aria-hidden="true">
        <circle cx="8" cy="8" r="2.5" stroke="currentColor"
stroke-width="1.5"/>
        <path d="M12.8 8a4.8 4.8 0 0
0-.26-1.57l1.48-1.48-1.06-1.06-1.48 1.48A4.8 4.8 0 0 0 9.9
4.48V3h-1.5v1.48a4.8 4.8 0 0 0-1.57.26L5.35 3.26 4.29
4.32l1.48 1.48A4.8 4.8 0 0 0 5.5 7.37H4v1.5h1.48c.07.54.22
1.06.44 1.57l-1.48 1.48 1.48 1.06 1.06 1.48-1.48-1.48c.51.22 1.03.37
1.57.44V15h1.5v-1.48c.54-.07 1.06-.22 1.57-.44l1.48 1.48
1.06-1.06-1.48-1.48c.22-.51.37-1.03.44-1.57H15v-1.5h-1.48z"
stroke="currentColor" stroke-width="1.5"/>
    </svg>
</button>
</div>
</header>

<!-- Connection Status Bar -->
<div id="connection-status" class="connection-bar"
role="status" aria-live="polite">
    <span class="status-dot" id="status-dot"></span>
    <span class="status-text" id="status-text">Connecting...</
span>
</div>

<!-- Main Content: Scrollable -->
<main class="popup-content" id="main-content">
    <!-- Empty state (shown when no torrents detected) -->
    <div id="empty-state" class="empty-state">
        <svg width="48" height="48" viewBox="0 0 48 48"
fill="none" class="empty-icon" aria-hidden="true">
            <path d="M24 4v8M24 36v8M4 24h8M36 24h8"
stroke="currentColor" stroke-width="2" stroke-
linecap="round"/>
            <circle cx="24" cy="24" r="8" stroke="currentColor"
stroke-width="2"/>
        </svg>
        <p class="empty-title" data-i18n="noTorrentsFound">No
torrents found</p>
        <p class="empty-subtitle" data-i18n="scanThisPage">Click
refresh to scan this page</p>
    </div>

    <!-- Torrent List (hidden initially) -->
    <div id="torrent-list-container" class="torrent-container"
style="display: none;">
        <div class="torrent-list-header">

```

```

        <label class="select-all-label">
            <input type="checkbox" id="select-all" aria-
label="Select all torrents" />
            <span id="torrent-count" data-i18n="torrentsFound">0
torrents</span>
        </label>
        <button id="send-selected" class="btn btn-primary btn-
sm" disabled>
            <span data-i18n="sendToBoba">Send to Boba</span>
        </button>
    </div>

    <ul id="torrent-list" class="torrent-list" role="listbox"
aria-label="Detected torrents">
        <!-- Dynamically populated -->
    </li></ul>

    <button id="send-all" class="btn btn-primary btn-full"
data-i18n="sendAllToBoba">
        Send All to Boba
    </button>
</div>

<!-- Sending progress state -->
<div id="sending-state" class="sending-state"
style="display: none;">
    <div class="progress-ring" role="progressbar" aria-
valuemin="0" aria-valuemax="100" aria-valuenow="0" aria-
label="Sending torrents">
        <svg class="progress-ring-svg" viewBox="0 0 100 100">
            <circle class="progress-track" cx="50" cy="50"
r="40" />
            <circle class="progress-fill" cx="50" cy="50" r="40"
id="progress-circle" />
        </svg>
        <span class="progress-percent" id="progress-
percent">0%</span>
    </div>
    <p class="sending-text" data-
i18n="sendingTorrents">Sending to Boba...</p>
    <p class="sending-detail" id="sending-detail"></p>
</div>

<!-- Success state -->
<div id="success-state" class="success-state"
style="display: none;">
    <div class="success-icon">
        <svg width="48" height="48" viewBox="0 0 48 48"
fill="none">
            <circle cx="24" cy="24" r="20" stroke="#10B981"
stroke-width="2"/>
            <path d="M16 24l6 10-12" stroke="#10B981" stroke-
width="2.5" stroke-linecap="round" stroke-
linejoin="round"/>

```

```

        </svg>
      </div>
      <p class="success-title" data-
        i18n="sentSuccessfully">Sent successfully!</p>
      <p class="success-detail" id="success-detail"></p>
      <button id="view-dashboard-btn" class="btn btn-secondary
        btn-full" data-i18n="viewInDashboard">
        View in Dashboard
      </button>
    </div>
  </main>

  <!-- Footer: Fixed -->
  <footer class="popup-footer">
    <span class="footer-version">Boba v1.0</span>
    <a href="#" id="open-dashboard" class="footer-link" data-
      i18n="openDashboard">Dashboard</a>
  </footer>
</div>
<script src="popup.js" type="module"></script>
</body>
</html>

```

## 1.3 Popup CSS (Theme-Aware)

```

/* popup/popup.css */

/* === CSS Reset === */
*, *::before, *::after {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

/* === CSS Variables === */
:root {
  /* Light mode (default) */
  --bg-primary: #ffffff;
  --bg-secondary: #f8f9fa;
  --bg-tertiary: #e8eae5;
  --text-primary: #202124;
  --text-secondary: #5f6368;
  --text-tertiary: #80868b;
  --border-color: #dadce0;
  --accent-color: #1a73e8;
  --accent-hover: #1557b0;
  --accent-light: #d2e3fc;
  --success-color: #10b981;
  --error-color: #ef4444;
  --warning-color: #f59e0b;
  --shadow: 0 1px 3px rgba(60, 64, 67, 0.08);
  --radius: 8px;
}

```

```

    --radius-sm: 4px;

    /* Sizing */
    --popup-width: 400px;
    --header-height: 48px;
    --footer-height: 36px;
    --connection-height: 28px;
}

@media (prefers-color-scheme: dark) {
    :root {
        --bg-primary: #202124;
        --bg-secondary: #292a2d;
        --bg-tertiary: #3c4043;
        --text-primary: #e8eaed;
        --text-secondary: #9aa0a6;
        --text-tertiary: #80868b;
        --border-color: #3c4043;
        --accent-color: #8ab4f8;
        --accent-hover: #aecbfa;
        --accent-light: #1a1c1e;
        --shadow: 0 1px 3px rgba(0, 0, 0, 0.3);
    }
}

/* === Base Styles === */
html, body {
    width: var(--popup-width);
    min-height: 400px;
    max-height: 580px;
    overflow: hidden; /* scroll happens in main */
}

body {
    font-family: -apple-system, BlinkMacSystemFont, 'Segoe UI',
        Roboto, Oxygen, Ubuntu, sans-serif;
    font-size: 14px;
    line-height: 1.5;
    color: var(--text-primary);
    background: var(--bg-primary);
}

#app {
    display: flex;
    flex-direction: column;
    height: 100vh;
    max-height: 580px;
}

/* === Header === */
.popup-header {
    display: flex;

```

```
    align-items: center;
    justify-content: space-between;
    height: var(--header-height);
    padding: 0 16px;
    background: var(--bg-secondary);
    border-bottom: 1px solid var(--border-color);
    flex-shrink: 0;
}

.header-left {
    display: flex;
    align-items: center;
    gap: 8px;
}

.header-icon {
    border-radius: 4px;
}

.header-title {
    font-size: 15px;
    font-weight: 600;
    color: var(--text-primary);
}

.header-actions {
    display: flex;
    gap: 4px;
}

.icon-btn {
    display: flex;
    align-items: center;
    justify-content: center;
    width: 32px;
    height: 32px;
    border: none;
    border-radius: var(--radius-sm);
    background: transparent;
    color: var(--text-secondary);
    cursor: pointer;
    transition: background 0.15s, color 0.15s;
}

.icon-btn:hover {
    background: var(--bg-tertiary);
    color: var(--text-primary);
}

.icon-btn:focus-visible {
    outline: 2px solid var(--accent-color);
    outline-offset: 1px;
}
```

```

}

/* === Connection Status Bar === */
.connection-bar {
  display: flex;
  align-items: center;
  gap: 8px;
  height: var(--connection-height);
  padding: 0 16px;
  background: var(--bg-secondary);
  border-bottom: 1px solid var(--border-color);
  font-size: 12px;
  color: var(--text-secondary);
  flex-shrink: 0;
}

.status-dot {
  width: 8px;
  height: 8px;
  border-radius: 50%;
  background: var(--warning-color);
  flex-shrink: 0;
  transition: background 0.3s;
}

.status-dot.connected { background: var(--success-color); }
.status-dot.disconnected { background: var(--error-color); }
.status-dot.connecting { background: var(--warning-color);
  animation: pulse 1.5s infinite; }

@keyframes pulse {
  0%, 100% { opacity: 1; }
  50% { opacity: 0.4; }
}

/* === Main Content === */
.popup-content {
  flex: 1;
  overflow-y: auto;
  overflow-x: hidden;
  padding: 12px;
}

.popup-content::-webkit-scrollbar {
  width: 6px;
}

.popup-content::-webkit-scrollbar-track {
  background: transparent;
}

.popup-content::-webkit-scrollbar-thumb {

```



```
    background: var(--bg-tertiary);
    border-radius: 3px;
}

/* === Empty State === */
.empty-state {
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    padding: 40px 20px;
    text-align: center;
    color: var(--text-secondary);
}

.empty-icon {
    margin-bottom: 16px;
    color: var(--text-tertiary);
}

.empty-title {
    font-size: 15px;
    font-weight: 500;
    color: var(--text-primary);
    margin-bottom: 4px;
}

.empty-subtitle {
    font-size: 13px;
    color: var(--text-secondary);
}

/* === Torrent List === */
.torrent-container {
    display: flex;
    flex-direction: column;
    gap: 12px;
}

.torrent-list-header {
    display: flex;
    align-items: center;
    justify-content: space-between;
    padding: 4px 4px;
}

.select-all-label {
    display: flex;
    align-items: center;
    gap: 8px;
    font-size: 13px;
    color: var(--text-secondary);
}
```

```

    cursor: pointer;
}

.select-all-label input[type="checkbox"] {
    width: 16px;
    height: 16px;
    accent-color: var(--accent-color);
    cursor: pointer;
}

.torrent-list {
    list-style: none;
    display: flex;
    flex-direction: column;
    gap: 6px;
    max-height: 340px;
    overflow-y: auto;
}

/* Individual torrent item */
.torrent-item {
    display: flex;
    align-items: flex-start;
    gap: 10px;
    padding: 10px 12px;
    border-radius: var(--radius);
    background: var(--bg-secondary);
    border: 1px solid var(--border-color);
    transition: border-color 0.15s, box-shadow 0.15s;
    cursor: pointer;
}

.torrent-item:hover {
    border-color: var(--accent-color);
    box-shadow: var(--shadow);
}

.torrent-item:focus-within {
    outline: 2px solid var(--accent-color);
    outline-offset: -1px;
}

.torrent-item.selected {
    border-color: var(--accent-color);
    background: var(--accent-light);
}

.torrent-checkbox {
    margin-top: 2px;
    width: 16px;
    height: 16px;
    accent-color: var(--accent-color);
}

```

```

    flex-shrink: 0;
    cursor: pointer;
}

.torrent-info {
    flex: 1;
    min-width: 0;
}

.torrent-name {
    font-size: 13px;
    font-weight: 500;
    color: var(--text-primary);
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
    display: block;
}

.torrent-meta {
    display: flex;
    gap: 8px;
    margin-top: 2px;
    font-size: 11px;
    color: var(--text-tertiary);
}

.torrent-size::before { content: attr(data-size); }
.torrent-type-magnet::before { content: 'MAGNET'; }
.torrent-type-torrent::before { content: '.TORRENT'; }

.torrent-type-magnet,
.torrent-type-torrent {
    padding: 1px 5px;
    border-radius: 3px;
    font-size: 10px;
    font-weight: 600;
    letter-spacing: 0.3px;
}

.torrent-type-magnet {
    background: #dbeafe;
    color: #1d4ed8;
}

.torrent-type-torrent {
    background: #d1fae5;
    color: #047857;
}

@media (prefers-color-scheme: dark) {
    .torrent-type-magnet { background: #1e3a5f; color: #93c5fd; }

```

```

.torrent-type-torrent { background: #064e3b; color: #6ee7b7; }
}

/* === Buttons === */
.btn {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 6px;
  padding: 9px 16px;
  border: none;
  border-radius: var(--radius-sm);
  font-size: 13px;
  font-weight: 500;
  cursor: pointer;
  transition: all 0.15s ease;
}

.btn-primary {
  background: var(--accent-color);
  color: #ffffff;
}

.btn-primary:hover:not(:disabled) {
  background: var(--accent-hover);
}

.btn-primary:disabled {
  opacity: 0.5;
  cursor: not-allowed;
}

.btn-secondary {
  background: var(--bg-tertiary);
  color: var(--text-primary);
  border: 1px solid var(--border-color);
}

.btn-secondary:hover {
  background: var(--border-color);
}

.btn-sm {
  padding: 6px 12px;
  font-size: 12px;
}

.btn-full {
  width: 100%;
  margin-top: 8px;
}

```

```
/* === Progress Ring === */
.sending-state {
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: center;
  padding: 40px 20px;
}

.progress-ring {
  position: relative;
  width: 100px;
  height: 100px;
}

.progress-ring-svg {
  width: 100%;
  height: 100%;
  transform: rotate(-90deg);
}

.progress-track {
  fill: none;
  stroke: var(--bg-tertiary);
  stroke-width: 6;
}

.progress-fill {
  fill: none;
  stroke: var(--accent-color);
  stroke-width: 6;
  stroke-linecap: round;
  stroke-dasharray: 251.33; /* 2 * PI * 40 */
  stroke-dashoffset: 251.33;
  transition: stroke-dashoffset 0.3s ease;
}

.progress-percent {
  position: absolute;
  top: 50%;
  left: 50%;
  transform: translate(-50%, -50%);
  font-size: 20px;
  font-weight: 600;
  color: var(--text-primary);
}

.sending-text {
  margin-top: 16px;
  font-size: 15px;
  font-weight: 500;
  color: var(--text-primary);
}
```

```

}

.sending-detail {
  margin-top: 4px;
  font-size: 13px;
  color: var(--text-secondary);
}

/* === Success State === */
.success-state {
  display: flex;
  flex-direction: column;
  align-items: center;
  padding: 40px 20px;
  text-align: center;
}

.success-icon {
  margin-bottom: 16px;
}

.success-title {
  font-size: 15px;
  font-weight: 600;
  color: var(--text-primary);
  margin-bottom: 4px;
}

.success-detail {
  font-size: 13px;
  color: var(--text-secondary);
  margin-bottom: 16px;
}

/* === Footer === */
.popup-footer {
  display: flex;
  align-items: center;
  justify-content: space-between;
  height: var(--footer-height);
  padding: 0 16px;
  background: var(--bg-secondary);
  border-top: 1px solid var(--border-color);
  font-size: 11px;
  color: var(--text-tertiary);
  flex-shrink: 0;
}

.footer-link {
  color: var(--accent-color);
  text-decoration: none;
  font-weight: 500;
}

```

```

}

.footer-link: hover {
  text-decoration: underline;
}

/* === Focus Styles === */
: focus-visible {
  outline: 2px solid var(--accent-color);
  outline-offset: 2px;
}

/* === Reduced Motion === */
@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after {
    animation-duration: 0.01ms !important;
    animation-iteration-count: 1 !important;
    transition-duration: 0.01ms !important;
  }
}

/* === High Contrast === */
@media (prefers-contrast: high) {
  .torrent-item {
    border-width: 2px;
  }
  .btn {
    border: 2px solid currentColor;
  }
}

```

## 1.4 Popup JavaScript

```

// popup/popup.js - Manifest V3, type="module"

// ===== State =====
let detectedTorrents = [];
let selectedIds = new Set();
let isSending = false;

// ===== DOM Elements =====
const els = {
  emptyState: document.getElementById('empty-state'),
  torrentContainer: document.getElementById('torrent-list-container'),
  torrentList: document.getElementById('torrent-list'),
  torrentCount: document.getElementById('torrent-count'),
  selectAll: document.getElementById('select-all'),
  sendSelected: document.getElementById('send-selected'),
  sendAll: document.getElementById('send-all'),
  sendingState: document.getElementById('sending-state'),
}

```

```

    successState: document.getElementById('success-state'),
    successDetail: document.getElementById('success-detail'),
    progressCircle: document.getElementById('progress-circle'),
    progressPercent: document.getElementById('progress-percent'),
    sendingDetail: document.getElementById('sending-detail'),
    statusDot: document.getElementById('status-dot'),
    statusText: document.getElementById('status-text'),
    refreshBtn: document.getElementById('refresh-btn'),
    settingsBtn: document.getElementById('settings-btn'),
    viewDashboardBtn: document.getElementById('view-dashboard-btn'),
    openDashboard: document.getElementById('open-dashboard'),
};

// ===== Utility: i18n helper =====
function t(key, substitutions) {
    return chrome.i18n.getMessage(key, substitutions) || key;
}

// ===== Initialize =====
document.addEventListener('DOMContentLoaded', async () => {
    // Load i18n strings
    document.querySelectorAll('[data-i18n]').forEach(el => {
        const key = el.dataset.i18n;
        el.textContent = t(key);
    });

    // Check connection to Boba server
    await checkConnection();

    // Request torrent scan from content script
    await scanPage();

    // Set up event listeners
    setupEventListeners();
});

function setupEventListeners() {
    els.refreshBtn.addEventListener('click', scanPage);

    els.settingsBtn.addEventListener('click', () => {
        chrome.runtime.openOptionsPage();
    });

    els.selectAll.addEventListener('change', (e) => {
        const checkboxes = els.torrentList.querySelectorAll('.torrent-checkbox');
        checkboxes.forEach(cb => {
            cb.checked = e.target.checked;
            const id = cb.dataset.id;
            e.target.checked ? selectedIds.add(id) :
                selectedIds.delete(id);
        });
    });
}

```



```

        updateTorrentItems();
        updateSendButton();
    });

    els.sendSelected.addEventListener('click', () =>
        sendTorrents(Array.from(selectedIds)));
    els.sendAll.addEventListener('click', () => {
        const allIds = detectedTorrents.map(t => t.id);
        sendTorrents(allIds);
    });

    els.viewDashboardBtn.addEventListener('click', openDashboard);
    els.openDashboard.addEventListener('click', (e) => {
        e.preventDefault();
        openDashboard();
    });
}

// ===== Connection Check =====
async function checkConnection() {
    try {
        const settings = await
            chrome.storage.sync.get('bobaServerUrl');
        const url = settings.bobaServerUrl || 'http://localhost:8080';

        updateConnectionStatus('connecting', t('connecting'));

        const response = await fetch(`${url}/api/health`, {
            method: 'GET',
            signal: AbortSignal.timeout(5000)
        });

        if (response.ok) {
            updateConnectionStatus('connected', t('connected'));
        } else {
            updateConnectionStatus('disconnected', t('serverError'));
        }
    } catch (err) {
        updateConnectionStatus('disconnected', t('notConnected'));
    }
}

function updateConnectionStatus(state, text) {
    els.statusDot.className = 'status-dot';
    els.statusDot.classList.add(state);
    els.statusText.textContent = text;
}

// ===== Page Scanning =====
async function scanPage() {
    updateConnectionStatus('connecting', t('scanning'));

```

```

try {
  const [tab] = await chrome.tabs.query({ active: true,
    currentWindow: true });
  if (!tab?.id) return;

  // Inject content script if not already injected
  const results = await chrome.scripting.executeScript({
    target: { tabId: tab.id },
    func: () => {
      // Scan for magnet links and .torrent links
      const magnets =
        Array.from(document.querySelectorAll('a[href^="magnet:"]'))
          .map(a => ({
            type: 'magnet',
            url: a.href,
            name: a.textContent.trim() || a.href.substring(0, 60)
          + '...',
            source: window.location.href
          }));

      const torrents =
        Array.from(document.querySelectorAll('a[href$=".torrent"]'))
          .map(a => ({
            type: 'torrent',
            url: a.href,
            name: a.textContent.trim() || a.href.split('/').pop(),
            source: window.location.href
          }));

      // Also check for common torrent tracker patterns in page
      text
      const magnetMatches = document.body.innerText.match(/
        magnet:\?xt=urn:[a-z0-9]+:[a-z0-9]+/gi) || [];

      return { magnets, torrents, magnetTextMatches:
        magnetMatches.length };
    }
  });

  const scanResult = results[0]?.result || { magnets: [],
    torrents: [], magnetTextMatches: 0 };

  // Merge and deduplicate
  const all = [...scanResult.magnets, ...scanResult.torrents];
  detectedTorrents = all.map((item, idx) => ({ ...item, id:
    `torrent-${idx}` }));

  renderTorrentList();

  // Update badge via background
  await chrome.runtime.sendMessage({
    type: 'UPDATE_BADGE',
    count: detectedTorrents.length
  });
}

```

```

    });

    } catch (err) {
        console.error('Scan failed:', err);
        showEmptyState(t('scanError'), t('tryAgain'));
    }
}

// ===== Render Torrent List =====
function renderTorrentList() {
    if (detectedTorrents.length === 0) {
        els.emptyState.style.display = 'flex';
        els.torrentContainer.style.display = 'none';
        return;
    }

    els.emptyState.style.display = 'none';
    els.torrentContainer.style.display = 'flex';
    els.torrentCount.textContent = t('torrentsFound',
        [String(detectedTorrents.length)]);

    els.torrentList.innerHTML = detectedTorrents.map(torrent => `
        <li class="torrent-item" data-id="$
            {torrent.id}" role="option" tabindex="0">
            <input type="checkbox" class="torrent-checkbox" data-id="$
                {torrent.id}"
                ${selectedIds.has(torrent.id) ? 'checked' : ''}
                aria-label="Select ${escapeHtml(torrent.name)}" />
            <div class="torrent-info">
                <span class="torrent-name" title="$
                    {escapeHtml(torrent.url)}">${escapeHtml(torrent.name)}</
                    span>
                <div class="torrent-meta">
                    <span class="torrent-type-${torrent.type}"></span>
                    ${torrent.size ? `<span class="torrent-size" data-
                        size="${torrent.size}"></span>` : ''}
                </div>
            </div>
        </li>
    `).join('');

    // Attach item-level event listeners
    els.torrentList.querySelectorAll('.torrent-item').forEach(item
        => {
            const id = item.dataset.id;

            item.addEventListener('click', (e) => {
                if (e.target.classList.contains('torrent-checkbox')) return;
                const cb = item.querySelector('.torrent-checkbox');
                cb.checked = !cb.checked;
                toggleSelection(id, cb.checked);
            });
        });
}

```

```

        item.addEventListener('keydown', (e) => {
            if (e.key === 'Enter' || e.key === ' ') {
                e.preventDefault();
                const cb = item.querySelector('.torrent-checkbox');
                cb.checked = !cb.checked;
                toggleSelection(id, cb.checked);
            }
        });
    });

    els.torrentList.querySelectorAll('.torrent-checkbox').forEach(cb => {
        cb.addEventListener('change', (e) => {
            toggleSelection(e.target.dataset.id, e.target.checked);
        });
    });
}

function toggleSelection(id, isSelected) {
    isSelected ? selectedIds.add(id) : selectedIds.delete(id);
    updateTorrentItems();
    updateSendButton();
}

function updateTorrentItems() {
    els.torrentList.querySelectorAll('.torrent-item').forEach(item
        => {
            const id = item.dataset.id;
            item.classList.toggle('selected', selectedIds.has(id));
        });

    // Update select-all checkbox state
    const allCount = detectedTorrents.length;
    const checkedCount = selectedIds.size;
    els.selectAll.checked = checkedCount > 0;
    els.selectAll.indeterminate = checkedCount > 0 && checkedCount
        < allCount;
}

function updateSendButton() {
    els.sendSelected.disabled = selectedIds.size === 0 || isSending;
    const count = selectedIds.size;
    els.sendSelected.innerHTML = count > 0
        ? `${t('sendToBoba')} (${count})`
        : t('sendToBoba');
}

function showEmptyState(title, subtitle) {
    els.emptyState.style.display = 'flex';
    els.emptyState.querySelector('.empty-title').textContent =
        title;
    els.emptyState.querySelector('.empty-subtitle').textContent =
        subtitle;
}

```

```

    els.torrentContainer.style.display = 'none';
}

function escapeHtml(text) {
    const div = document.createElement('div');
    div.textContent = text;
    return div.innerHTML;
}

// ===== Send Torrents =====
async function sendTorrents(ids) {
    if (isSending) return;
    isSending = true;

    const toSend = detectedTorrents.filter(t => ids.includes(t.id));

    // Show sending UI
    els.emptyState.style.display = 'none';
    els.torrentContainer.style.display = 'none';
    els.sendingState.style.display = 'flex';
    els.successState.style.display = 'none';

    updateProgress(0);

    try {
        const settings = await
            chrome.storage.sync.get(['bobaServerUrl', 'apiToken']);
        const url = settings.bobaServerUrl || 'http://localhost:8080';
        const token = settings.apiToken || '';

        for (let i = 0; i < toSend.length; i++) {
            const torrent = toSend[i];
            const percent = Math.round(((i + 0.5) / toSend.length) *
                100);
            updateProgress(percent);
            els.sendingDetail.textContent = t('sendingItem', [String(i
                + 1), String(toSend.length),
                escapeHtml(torrent.name.substring(0, 40))]);

            await fetch(`${url}/api/torrents/add`, {
                method: 'POST',
                headers: {
                    'Content-Type': 'application/json',
                    ...(token && { 'Authorization': `Bearer ${token}` })
                },
                body: JSON.stringify({
                    url: torrent.url,
                    name: torrent.name,
                    type: torrent.type,
                    source: torrent.source
                }),
                signal: AbortSignal.timeout(15000)
            });
        }
    } catch (error) {
        console.error('Error sending torrent:', error);
    }
}

```

```

    });

    updateProgress(Math.round(((i + 1) / toSend.length) * 100));
  }

  // Show success
  els.sendingState.style.display = 'none';
  els.successState.style.display = 'flex';
  els.successDetail.textContent = t('sentDetail',
    [String(toSend.length)]);

  // Show notification
  chrome.runtime.sendMessage({
    type: 'SHOW_NOTIFICATION',
    title: t('sentTitle'),
    message: t('sentMessage', [String(toSend.length)])
  });

} catch (err) {
  console.error('Send failed:', err);
  els.sendingDetail.textContent = t('sendError');
  els.sendingDetail.style.color = 'var(--error-color)';
} finally {
  isSending = false;
}
}

function updateProgress(percent) {
  const circumference = 2 * Math.PI * 40; // r=40
  const offset = circumference - (percent / 100) * circumference;
  els.progressCircle.style.strokeDashoffset = offset;
  els.progressPercent.textContent = `${percent}%`;
  els.progressCircle.parentElement.setAttribute('aria-valuenow',
    String(percent));
}

function openDashboard() {
  chrome.storage.sync.get('bobaServerUrl', (result) => {
    const url = result.bobaServerUrl || 'http://localhost:8080';
    chrome.tabs.create({ url });
  });
}

```

---

## 2. Options Page

### 2.1 Options Page HTML

```

<!-- options/options.html -->
<!DOCTYPE html>

```

```

<html lang="en" data-theme="auto">
<head>
  <meta charset="UTF-8" />
  <meta name="color-scheme" content="light dark" />
  <meta name="viewport" content="width=device-width, initial-
    scale=1.0" />
  <title>Boba Settings</title>
  <link rel="stylesheet" href="options.css" />
</head>
<body>
  <div class="options-container">
    <header class="options-header">
      
      <h1 data-il8n="optionsTitle">Boba Settings</h1>
    </header>

    <main class="options-content">
      <!-- Server Connection Section -->
      <section class="options-section" aria-labelledby="server-
        heading">
        <h2 id="server-heading" class="section-title">
          <span class="section-icon">
            <svg width="20" height="20" viewBox="0 0 24 24"
              fill="none" stroke="currentColor" stroke-width="2">
              <rect x="2" y="2" width="20" height="8" rx="2"/>
              <rect x="2" y="14" width="20" height="8" rx="2"/>
              <line x1="6" y1="6" x2="6" y2="6"/>
              <line x1="6" y1="18" x2="6" y2="18"/>
            </svg>
          </span>
          <span data-il8n="serverConnection">Server Connection</
            span>
        </h2>
        <p class="section-desc" data-il8n="serverConnectionDesc">
          Configure your Boba server URL and API credentials.
        </p>

        <div class="form-group">
          <label for="server-url" class="form-label" data-
            il8n="serverUrlLabel">Boba Server URL</label>
          <input
            type="url"
            id="server-url"
            class="form-input"
            placeholder="http://localhost:8080"
            required
            aria-describedby="server-url-help"
          />
          <p id="server-url-help" class="form-help" data-
            il8n="serverUrlHelp">
            The URL where your Boba server is running. Include the
            port if not using standard ports.
          </p>
        </div>
      </main>
    </div>
  </body>
</html>

```

```

    </p>
  </div>

  <div class="form-group">
    <label for="api-token" class="form-label" data-
      il8n="apiTokenLabel">API Token (Optional)</label>
    <div class="input-group">
      <input
        type="password"
        id="api-token"
        class="form-input"
        placeholder="Enter your API token"
        aria-describedby="api-token-help"
      />
      <button type="button" id="toggle-token" class="input-
        btn" aria-label="Show token">
        <svg width="18" height="18" viewBox="0 0 24 24"
          fill="none" stroke="currentColor" stroke-width="2">
          <path d="M1 12s4-8 11-8 11 8 11 8-4 8-11
            8-11-8-11-8z"/>
          <circle cx="12" cy="12" r="3"/>
        </svg>
      </button>
    </div>
    <p id="api-token-help" class="form-help" data-
      il8n="apiTokenHelp">
      If your Boba server requires authentication, enter
      your API token here.
    </p>
  </div>

  <div class="form-actions">
    <button type="button" id="test-connection" class="btn
      btn-secondary">
      <span data-il8n="testConnection">Test Connection</
        span>
    </button>
    <span id="connection-result" class="connection-result"
      role="status" aria-live="polite"></span>
  </div>
</section>

<!-- Download Settings Section -->
<section class="options-section" aria-labelledby="download-
  heading">
  <h2 id="download-heading" class="section-title">
    <span class="section-icon">
      <svg width="20" height="20" viewBox="0 0 24 24"
        fill="none" stroke="currentColor" stroke-width="2">
        <path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/
        >
        <polyline points="7 10 12 15 17 10"/>
        <line x1="12" y1="15" x2="12" y2="3"/>
      </svg>
    
```



```

    </span>
    <span data-i18n="downloadSettings">Download Settings</
span>
</h2>

<div class="form-group">
  <label class="form-label" data-
i18n="defaultCategory">Default Category</label>
  <select id="default-category" class="form-select">
    <option value="" data-i18n="categoryNone">None</
option>
    <option value="movies" data-
i18n="categoryMovies">Movies</option>
    <option value="tv" data-i18n="categoryTV">TV Shows</
option>
    <option value="music" data-
i18n="categoryMusic">Music</option>
    <option value="software" data-
i18n="categorySoftware">Software</option>
    <option value="other" data-
i18n="categoryOther">Other</option>
  </select>
</div>

<div class="form-group">
  <label class="form-label toggle-label">
    <input type="checkbox" id="auto-start" class="toggle-
input" />
    <span class="toggle-switch" aria-hidden="true"></span>
    <span class="toggle-text" data-i18n="autoStart">Auto-
start downloads</span>
  </label>
  <p class="form-help" data-i18n="autoStartHelp">
    Automatically start downloading torrents when sent to
    Boba.
  </p>
</div>

<div class="form-group">
  <label class="form-label toggle-label">
    <input type="checkbox" id="notifications"
class="toggle-input" checked />
    <span class="toggle-switch" aria-hidden="true"></span>
    <span class="toggle-text" data-
i18n="enableNotifications">Enable notifications</span>
  </label>
  <p class="form-help" data-i18n="notificationsHelp">
    Show browser notifications for download events.
  </p>
</div>
</section>

<!-- Behavior Section -->

```

```

<section class="options-section" aria-labelledby="behavior-
  heading">
  <h2 id="behavior-heading" class="section-title">
    <span class="section-icon">
      <svg width="20" height="20" viewBox="0 0 24 24"
        fill="none" stroke="currentColor" stroke-width="2">
        <circle cx="12" cy="12" r="3"/>
        <path d="M19.4 15a1.65 1.65 0 0 0 .33 1.82l.06.06a2
          2 0 0 1 0 2.83 2 2 0 0 1-2.83 0l-.06-.06a1.65 1.65 0 0
          0-1.82-.33 1.65 1.65 0 0 0-1 1.51V21a2 2 0 0 1-2 2 2 2 0 0
          1-2-2v-.09A1.65 1.65 0 0 0 9 19.4a1.65 1.65 0 0
          0-1.82.33l-.06.06a2 2 0 0 1-2.83 0 2 2 0 0 1
          0-2.83l.06-.06a1.65 1.65 0 0 0 .33-1.82 1.65 1.65 0 0
          0-1.51-1H3a2 2 0 0 1-2-2 2 2 0 0 1 2-2h.09A1.65 1.65 0 0 0
          4.6 9a1.65 1.65 0 0 0-.33-1.82l-.06-.06a2 2 0 0 1 0-2.83 2
          2 0 0 1 2.83 0l.06.06a1.65 1.65 0 0 0 1.82.33H9a1.65 1.65
          0 0 0 1-1.51V3a2 2 0 0 1 2-2 2 2 0 0 1 2 2v.09a1.65 1.65 0
          0 0 1 1.51 1.65 1.65 0 0 0 1.82-.33l.06-.06a2 2 0 0 1 2.83
          0 2 2 0 0 1 0 2.83l-.06.06a1.65 1.65 0 0 0-.33 1.82V9a1.65
          1.65 0 0 0 1.51 1H21a2 2 0 0 1 2 2 2 2 0 0 1-2 2h-.09a1.65
          1.65 0 0 0-1.51 1z"/>
        </svg>
      </span>
      <span data-il8n="behavior">Behavior</span>
    </h2>

    <div class="form-group">
      <label class="form-label" data-il8n="scanMode">Scan
        Mode</label>
      <select id="scan-mode" class="form-select">
        <option value="manual" data-il8n="scanManual">Manual
          (click to scan)</option>
        <option value="auto" data-il8n="scanAuto">Automatic
          (on page load)</option>
      </select>
    </div>

    <div class="form-group">
      <label class="form-label toggle-label">
        <input type="checkbox" id="context-menu"
          class="toggle-input" checked="" />
        <span class="toggle-switch" aria-hidden="true"></span>
        <span class="toggle-text" data-
          il8n="contextMenu">Show context menu items</span>
      </label>
    </div>
  </section>

  <!-- Danger Zone -->
  <section class="options-section danger-section" aria-
    labelledby="danger-heading">
    <h2 id="danger-heading" class="section-title danger-
      title" data-il8n="dangerZone">Danger Zone</h2>
    <button type="button" id="reset-settings" class="btn btn-
      danger">

```

```

        <span data-il8n="resetSettings">Reset All Settings</
        span>
    </button>
</section>
</main>

<footer class="options-footer">
    <button type="button" id="save-settings" class="btn btn-
        primary btn-lg">
        <span data-il8n="saveSettings">Save Settings</span>
    </button>
    <span id="save-status" class="save-status" role="status"
        aria-live="polite"></span>
</footer>
</div>
<script src="options.js" type="module"></script>
</body>
</html>

```

## 2.2 Options Page CSS

```

/* options/options.css */
*, *::before, *::after {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
}

:root {
    --bg-primary: #ffffff;
    --bg-secondary: #f8f9fa;
    --text-primary: #202124;
    --text-secondary: #5f6368;
    --text-tertiary: #80868b;
    --border-color: #dadce0;
    --accent: #1a73e8;
    --accent-hover: #1557b0;
    --success: #10b981;
    --error: #ef4444;
    --danger: #dc2626;
    --radius: 8px;
}

@media (prefers-color-scheme: dark) {
    :root {
        --bg-primary: #202124;
        --bg-secondary: #292a2d;
        --text-primary: #e8eaed;
        --text-secondary: #9aa0a6;
        --text-tertiary: #80868b;
        --border-color: #3c4043;
        --accent: #8ab4f8;
    }
}

```

```

        --accent-hover: #aecbfa;
        --danger: #ef4444;
    }
}

body {
    font-family: -apple-system, BlinkMacSystemFont, 'Segoe UI',
        Roboto, sans-serif;
    font-size: 14px;
    line-height: 1.5;
    color: var(--text-primary);
    background: var(--bg-secondary);
    min-height: 100vh;
}

.options-container {
    max-width: 640px;
    margin: 0 auto;
    padding: 24px 16px;
}

/* Header */
.options-header {
    display: flex;
    align-items: center;
    gap: 12px;
    margin-bottom: 24px;
    padding-bottom: 16px;
    border-bottom: 1px solid var(--border-color);
}

.options-header h1 {
    font-size: 22px;
    font-weight: 600;
}

/* Sections */
.options-section {
    background: var(--bg-primary);
    border: 1px solid var(--border-color);
    border-radius: var(--radius);
    padding: 24px;
    margin-bottom: 16px;
}

.section-title {
    display: flex;
    align-items: center;
    gap: 8px;
    font-size: 16px;
    font-weight: 600;
    margin-bottom: 8px;
}

```

```

}

.section-desc {
  font-size: 13px;
  color: var(--text-secondary);
  margin-bottom: 16px;
}

.section-icon {
  color: var(--text-tertiary);
}

/* Form Elements */
.form-group {
  margin-bottom: 16px;
}

.form-group:last-child {
  margin-bottom: 0;
}

.form-label {
  display: block;
  font-size: 13px;
  font-weight: 500;
  color: var(--text-primary);
  margin-bottom: 6px;
}

.form-input,
.form-select {
  width: 100%;
  padding: 10px 12px;
  border: 1px solid var(--border-color);
  border-radius: var(--radius);
  background: var(--bg-primary);
  color: var(--text-primary);
  font-size: 14px;
  transition: border-color 0.15s, box-shadow 0.15s;
}

.form-input:focus,
.form-select:focus {
  outline: none;
  border-color: var(--accent);
  box-shadow: 0 0 0 3px rgba(26, 115, 232, 0.15);
}

@media (prefers-color-scheme: dark) {
  .form-input:focus,
  .form-select:focus {
    box-shadow: 0 0 0 3px rgba(138, 180, 248, 0.15);
  }
}

```

```

    }
}

.form-help {
    font-size: 12px;
    color: var(--text-secondary);
    margin-top: 4px;
}

/* Input Group (for password show/hide) */
.input-group {
    display: flex;
    gap: 0;
}

.input-group .form-input {
    border-radius: var(--radius) 0 0 var(--radius);
    border-right: none;
}

.input-btn {
    padding: 10px 12px;
    border: 1px solid var(--border-color);
    border-radius: 0 var(--radius) var(--radius) 0;
    background: var(--bg-secondary);
    color: var(--text-secondary);
    cursor: pointer;
    transition: background 0.15s;
}

.input-btn:hover {
    background: var(--border-color);
}

/* Toggle Switch */
.toggle-label {
    display: flex !important;
    align-items: center;
    gap: 10px;
    cursor: pointer;
}

.toggle-input {
    position: absolute;
    opacity: 0;
    width: 0;
    height: 0;
}

.toggle-switch {
    width: 40px;
    height: 22px;

```

```

    background: var(--border-color);
    border-radius: 11px;
    position: relative;
    transition: background 0.2s;
    flex-shrink: 0;
}

.toggle-switch::after {
    content: '';
    position: absolute;
    width: 18px;
    height: 18px;
    border-radius: 50%;
    background: #fff;
    top: 2px;
    left: 2px;
    transition: transform 0.2s;
    box-shadow: 0 1px 3px rgba(0,0,0,0.2);
}

.toggle-input:checked + .toggle-switch {
    background: var(--accent);
}

.toggle-input:checked + .toggle-switch::after {
    transform: translateX(18px);
}

.toggle-input:focus-visible + .toggle-switch {
    outline: 2px solid var(--accent);
    outline-offset: 2px;
}

.toggle-text {
    font-size: 13px;
}

/* Buttons */
.btn {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    gap: 6px;
    padding: 10px 20px;
    border: none;
    border-radius: var(--radius);
    font-size: 14px;
    font-weight: 500;
    cursor: pointer;
    transition: all 0.15s;
}

```

```
.btn-primary {
  background: var(--accent);
  color: #fff;
}

.btn-primary:hover {
  background: var(--accent-hover);
}

.btn-secondary {
  background: var(--bg-secondary);
  color: var(--text-primary);
  border: 1px solid var(--border-color);
}

.btn-secondary:hover {
  background: var(--border-color);
}

.btn-danger {
  background: transparent;
  color: var(--danger);
  border: 1px solid var(--danger);
}

.btn-danger:hover {
  background: var(--danger);
  color: #fff;
}

.btn-lg {
  padding: 12px 32px;
  font-size: 15px;
}

/* Danger Section */
.danger-section {
  border-color: var(--danger);
}

.danger-title {
  color: var(--danger);
}

/* Form Actions */
.form-actions {
  display: flex;
  align-items: center;
  gap: 12px;
  margin-top: 16px;
}
```



```

.connection-result {
  font-size: 13px;
  font-weight: 500;
}

.connection-result.success { color: var(--success); }
.connection-result.error { color: var(--error); }

/* Footer */
.options-footer {
  display: flex;
  align-items: center;
  gap: 12px;
  padding-top: 8px;
}

.save-status {
  font-size: 13px;
  color: var(--text-secondary);
}

.save-status.success { color: var(--success); }

/* Focus */
:focus-visible {
  outline: 2px solid var(--accent);
  outline-offset: 2px;
}

@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after {
    transition-duration: 0.01ms !important;
  }
}

```

## 2.3 Options Page JavaScript

```

// options/options.js - Manifest V3, type="module"

// ===== Default Settings =====
const DEFAULT_SETTINGS = {
  bobaServerUrl: 'http://localhost:8080',
  apiToken: '',
  defaultCategory: '',
  autoStart: false,
  notifications: true,
  scanMode: 'manual',
  contextMenu: true
};

// ===== DOM Elements =====

```

```

const els = {
  serverUrl: document.getElementById('server-url'),
  apiToken: document.getElementById('api-token'),
  toggleToken: document.getElementById('toggle-token'),
  testConnection: document.getElementById('test-connection'),
  connectionResult: document.getElementById('connection-result'),
  defaultCategory: document.getElementById('default-category'),
  autoStart: document.getElementById('auto-start'),
  notifications: document.getElementById('notifications'),
  scanMode: document.getElementById('scan-mode'),
  contextMenu: document.getElementById('context-menu'),
  resetSettings: document.getElementById('reset-settings'),
  saveSettings: document.getElementById('save-settings'),
  saveStatus: document.getElementById('save-status')
};

// ===== Load Settings =====
document.addEventListener('DOMContentLoaded', async () => {
  // Apply i18n
  document.querySelectorAll('[data-i18n]').forEach(el => {
    const key = el.dataset.i18n;
    const msg = chrome.i18n.getMessage(key);
    if (msg) el.textContent = msg;
  });

  // Load saved settings
  const stored = await
    chrome.storage.sync.get(Object.keys(DEFAULT_SETTINGS));
  const settings = { ...DEFAULT_SETTINGS, ...stored };

  // Apply to form
  applySettingsToForm(settings);

  // Set up listeners
  setupListeners();

  // Auto-save on change (debounced)
  setupAutoSave();
});

function applySettingsToForm(settings) {
  els.serverUrl.value = settings.bobaServerUrl || '';
  els.apiToken.value = settings.apiToken || '';
  els.defaultCategory.value = settings.defaultCategory || '';
  els.autoStart.checked = settings.autoStart;
  els.notifications.checked = settings.notifications;
  els.scanMode.value = settings.scanMode;
  els.contextMenu.checked = settings.contextMenu;
}

function getSettingsFromForm() {
  return {

```

```

    bobaServerUrl: els.serverUrl.value.trim() ||
        DEFAULT_SETTINGS.bobaServerUrl,
    apiToken: els.apiToken.value.trim(),
    defaultCategory: els.defaultCategory.value,
    autoStart: els.autoStart.checked,
    notifications: els.notifications.checked,
    scanMode: els.scanMode.value,
    contextMenu: els.contextMenu.checked
};
}

function setupListeners() {
    // Toggle password visibility
    els.toggleToken.addEventListener('click', () => {
        const isPassword = els.apiToken.type === 'password';
        els.apiToken.type = isPassword ? 'text' : 'password';
        els.toggleToken.setAttribute('aria-label', isPassword ? 'Hide
            token' : 'Show token');
    });

    // Test Connection
    els.testConnection.addEventListener('click', async () => {
        els.connectionResult.textContent = '';
        const url = els.serverUrl.value.trim() ||
            DEFAULT_SETTINGS.bobaServerUrl;

        try {
            els.testConnection.disabled = true;
            els.connectionResult.textContent =
                chrome.i18n.getMessage('testing') || 'Testing...';
            els.connectionResult.className = 'connection-result';

            const response = await fetch(`${url}/api/health`, {
                signal: AbortSignal.timeout(8000)
            });

            if (response.ok) {
                els.connectionResult.textContent =
                    chrome.i18n.getMessage('connectionSuccess') ||
                    'Connected!';
                els.connectionResult.className = 'connection-result
                    success';
            } else {
                els.connectionResult.textContent =
                    chrome.i18n.getMessage('connectionFailed') || 'Server
                    error';
                els.connectionResult.className = 'connection-result
                    error';
            }
        } catch (err) {
            els.connectionResult.textContent =
                chrome.i18n.getMessage('connectionError') || 'Cannot
                connect';
            els.connectionResult.className = 'connection-result error';
        }
    });
}

```

```

    } finally {
      els.testConnection.disabled = false;
    }
  });

  // Reset Settings
  els.resetSettings.addEventListener('click', async () => {
    const confirmed =
      confirm(chrome.i18n.getMessage('resetConfirm') || 'Reset
      all settings?');
    if (confirmed) {
      await chrome.storage.sync.clear();
      applySettingsToForm(DEFAULT_SETTINGS);
      showSaveStatus('reset', 'Settings reset');
    }
  });

  // Save Button
  els.saveSettings.addEventListener('click', saveSettings);
}

let saveTimeout;
function setupAutoSave() {
  // Auto-save 500ms after any input change
  const inputs = document.querySelectorAll('input, select,
  textarea');
  inputs.forEach(input => {
    input.addEventListener('change', () => {
      clearTimeout(saveTimeout);
      saveTimeout = setTimeout(saveSettings, 500);
    });
  });
}

async function saveSettings() {
  try {
    const settings = getSettingsFromForm();
    await chrome.storage.sync.set(settings);
    showSaveStatus('success', chrome.i18n.getMessage('saved') ||
    'Settings saved');

    // Notify background that settings changed
    chrome.runtime.sendMessage({ type: 'SETTINGS_CHANGED',
    settings });
  } catch (err) {
    console.error('Save failed:', err);
    showSaveStatus('error', chrome.i18n.getMessage('saveFailed')
    || 'Save failed');
  }
}

function showSaveStatus(type, message) {
  els.saveStatus.textContent = message;
}

```

```

els.saveStatus.className = 'save-status';
if (type === 'success') els.saveStatus.classList.add('success');

setTimeout(() => {
  els.saveStatus.textContent = '';
}, 3000);
}

```

## 2.4 Storage Architecture

Claim: chrome.storage.sync should be used for settings (100KB total, 8KB/item). chrome.storage.local for larger data. chrome.storage.session for temporary MV3 state.

Source: Chrome Developer Docs

URL: <https://developer.chrome.com/docs/extensions/reference/api/storage>

Date: 2026-05-12

Excerpt: "storage.sync: data synced across devices, quota ~100KB total, 8KB per item... storage.local: device-specific, 10MB limit... storage.session: current browser session, 10MB, MV3 only"

Context: Use sync for user preferences, local for cached torrent lists, session for popup state

Confidence: high

Claim: The chrome.storage API preserves data types automatically - no JSON.parse/stringify needed. Supports monitoring changes via onChanged listener.

Source: Chrome Developer Docs / Reintech

URL: <https://developer.chrome.com/docs/extensions/reference/api/storage>

Date: 2026-05-12

Excerpt: "Unlike localStorage, chrome.storage preserves data types automatically... chrome.storage.onChanged.addListener((changes, namespace) => { ... })"

Context: Background script listens to settings changes to reconfigure context menus, icon state

Confidence: high

## 3. Context Menus

### 3.1 Context Menu Registration

*// background/contextMenus.js - Service Worker*

```

const CONTEXT_MENU_ITEMS = {
  SEND_TO_BOBA_LINK: 'send-to-boba-link',
  SEND_TO_BOBA_PAGE: 'send-to-boba-page',
  SCAN_THIS_PAGE: 'scan-this-page',

```

```

    SEPARATOR_1: 'separator-1',
    OPEN_DASHBOARD: 'open-dashboard'
};

/**
 * Initialize all context menu items on install.
 * Also call this when settings change (e.g., context menu
 *   preference toggled).
 */
export async function initContextMenus() {
    // Clear existing
    await chrome.contextMenus.removeAll();

    // Check if context menus are enabled in settings
    const settings = await chrome.storage.sync.get('contextMenu');
    if (settings.contextMenu === false) return;

    // --- "Send to Boba" on links ---
    // Magnet links
    chrome.contextMenus.create({
        id: CONTEXT_MENU_ITEMS.SEND_TO_BOBA_LINK,
        title: chrome.i18n.getMessage('sendToBoba') || 'Send to Boba',
        contexts: ['link'],
        targetUrlPatterns: [
            'magnet:*', // magnet links
            '*://*/*.torrent', // .torrent file links
            '*://*/*.torrent?*' // .torrent with query params
        ]
    });

    // --- "Scan this page" on page context ---
    chrome.contextMenus.create({
        id: CONTEXT_MENU_ITEMS.SCAN_THIS_PAGE,
        title: chrome.i18n.getMessage('scanThisPage') || 'Scan for
        torrents',
        contexts: ['page', 'action']
    });

    // --- Separator ---
    chrome.contextMenus.create({
        id: CONTEXT_MENU_ITEMS.SEPARATOR_1,
        type: 'separator',
        contexts: ['page', 'link']
    });

    // --- Open Dashboard ---
    chrome.contextMenus.create({
        id: CONTEXT_MENU_ITEMS.OPEN_DASHBOARD,
        title: chrome.i18n.getMessage('openDashboard') || 'Open Boba
        Dashboard',
        contexts: ['page', 'action']
    });
}

```

```

}

/**
 * Handle context menu clicks
 */
chrome.contextMenus.onClicked.addListener(async (info, tab) => {
  switch (info.menuItemId) {
    case CONTEXT_MENU_ITEMS.SEND_TO_BOBA_LINK:
      if (info.linkUrl) {
        await sendUrlToBoba(info.linkUrl, tab);
      }
      break;

    case CONTEXT_MENU_ITEMS.SCAN_THIS_PAGE:
      await chrome.tabs.sendMessage(tab.id, { type:
        'SCAN_REQUEST' });
      break;

    case CONTEXT_MENU_ITEMS.OPEN_DASHBOARD:
      const settings = await
        chrome.storage.sync.get('bobaServerUrl');
      const url = settings.bobaServerUrl || 'http://
        localhost:8080';
      chrome.tabs.create({ url });
      break;
  }
});

async function sendUrlToBoba(url, tab) {
  try {
    const settings = await
      chrome.storage.sync.get(['bobaServerUrl', 'apiToken',
        'defaultCategory', 'autoStart']);
    const serverUrl = settings.bobaServerUrl || 'http://
      localhost:8080';

    // Determine type
    const type = url.startsWith('magnet:') ? 'magnet' : 'torrent';

    const response = await fetch(`${serverUrl}/api/torrents/add`,
      {
        method: 'POST',
        headers: {
          'Content-Type': 'application/json',
          ...(settings.apiToken && { 'Authorization': `Bearer $
            {settings.apiToken}` })
        },
        body: JSON.stringify({
          url,
          type,
          source: tab.url,
          category: settings.defaultCategory || undefined,
          autoStart: settings.autoStart ?? false
        })
      }
    );
  } catch (error) {
    console.error('Error sending URL to Boba:', error);
  }
}

```

```

    })
  });

  if (response.ok) {
    showNotification(
      chrome.i18n.getMessage('sentTitle') || 'Sent to Boba',
      chrome.i18n.getMessage('linkSent') || `Added: $
        {url.substring(0, 60)}...`
    );
    // Update badge
    const current = await chrome.action.getBadgeText({});
    const count = parseInt(current) || 0;
    await chrome.action.setBadgeText({ text: String(count +
      1) });
  } else {
    showNotification(
      chrome.i18n.getMessage('sendFailed') || 'Failed to send',
      `Server returned ${response.status}`,
      'error'
    );
  }
} catch (err) {
  console.error('Failed to send:', err);
  showNotification(
    chrome.i18n.getMessage('sendFailed') || 'Failed to send',
    err.message,
    'error'
  );
}
}

function showNotification(title, message, type = 'basic') {
  chrome.notifications.create({
    type: type === 'progress' ? 'progress' : 'basic',
    iconUrl: chrome.runtime.getURL('icons/icon-128.png'),
    title,
    message,
    ...(type === 'progress' ? { progress: 50 } : {})
  });
}

// Re-init context menus when settings change
chrome.storage.onChanged.addListener((changes, area) => {
  if (area === 'sync' && changes.contextMenu) {
    initContextMenus();
  }
});

// Also init on install/update
chrome.runtime.onInstalled.addListener(() => {
  initContextMenus();
});

```



## 3.2 Context Menu API Reference

Claim: Context menus require "contextMenus" permission in manifest. Items can target specific URL patterns via `targetUrlPatterns`. Types include "normal", "checkbox", "radio", "separator".

Source: Chrome Developer Docs

URL: <https://developer.chrome.com/docs/extensions/reference/api/contextMenus>

Date: 2026-05-15

Excerpt: "contexts: ['link'] with targetUrlPatterns: ['magnet:\*', '\*://\*/\*.torrent']... The onclick property is deprecated in MV3 event pages; use `chrome.contextMenus.onClicked.addListener` instead"

Context: MV3 best practice is to use the event listener pattern, not inline onclick

Confidence: high

Claim: If an extension creates more than one context menu item visible at once, Chrome automatically collapses them into a single parent menu labeled with the extension name.

Source: MDN Web Docs

URL: <https://developer.mozilla.org/en-US/docs/Mozilla/Add-ons/WebExtensions/API/menus>

Date: 2025-07-17

Excerpt: "If you have created more than one context menu item... then the items will be placed in a submenu. The submenu's parent will be labeled with the name of the extension."

Context: Keep menu items focused and minimal to avoid submenu nesting

Confidence: high

---

## 4. chrome.action API

### 4.1 Badge & Icon State Management

*// background/iconState.js - Service Worker*

```
/**
 * Icon state definitions and management
 * States: idle, detecting, sending, error, connected,
 *         disconnected
 */

const ICON_STATES = {
  IDLE: {
    badgeText: '',
    badgeColor: [128, 128, 128, 255], // gray
    iconPath: {
```

```

        '16': 'icons/icon-16.png',
        '32': 'icons/icon-32.png',
        '48': 'icons/icon-48.png'
    },
    title: 'Boba - Click to scan'
},
DETECTING: {
    badgeText: '...',
    badgeColor: [245, 166, 35, 255],    // orange
    iconPath: {
        '16': 'icons/icon-detecting-16.png',
        '32': 'icons/icon-detecting-32.png'
    },
    title: 'Boba - Scanning...'
},
FOUND: {
    badgeColor: [26, 115, 232, 255],    // blue
    iconPath: {
        '16': 'icons/icon-16.png',
        '32': 'icons/icon-32.png'
    }
},
SENDING: {
    badgeText: '...',
    badgeColor: [245, 166, 35, 255],    // orange
    iconPath: {
        '16': 'icons/icon-sending-16.png',
        '32': 'icons/icon-sending-32.png'
    },
    title: 'Boba - Sending...'
},
ERROR: {
    badgeText: '!',
    badgeColor: [239, 68, 68, 255],    // red
    iconPath: {
        '16': 'icons/icon-error-16.png',
        '32': 'icons/icon-error-32.png'
    },
    title: 'Boba - Connection error'
},
CONNECTED: {
    badgeColor: [16, 185, 129, 255],    // green
    title: 'Boba - Connected'
},
DISCONNECTED: {
    badgeText: 'X',
    badgeColor: [239, 68, 68, 255],    // red
    iconPath: {
        '16': 'icons/icon-offline-16.png',
        '32': 'icons/icon-offline-32.png'
    },
    title: 'Boba - Disconnected'
}

```

```

    }
};

/**
 * Set the badge with detected torrent count
 */
export async function updateBadgeCount(count) {
    if (count === 0) {
        await chrome.action.setBadgeText({ text: '' });
        await chrome.action.setTitle({ title:
            ICON_STATES.IDLE.title });
    } else {
        // Badge text max ~4 chars
        const text = count > 999 ? '999+' : String(count);
        await chrome.action.setBadgeText({ text });
        await chrome.action.setBadgeBackgroundColor({ color:
            ICON_STATES.FOUND.badgeColor });
        await chrome.action.setTitle({ title: `Boba - ${count}
            torrents found` });
    }
}

/**
 * Set full icon state (for state-based icons)
 */
export async function setIconState(stateKey) {
    const state = ICON_STATES[stateKey];
    if (!state) return;

    if (state.badgeText !== undefined) {
        await chrome.action.setBadgeText({ text: state.badgeText });
    }
    if (state.badgeColor) {
        await chrome.action.setBadgeBackgroundColor({ color:
            state.badgeColor });
    }
    if (state.iconPath) {
        await chrome.action.setIcon({ path: state.iconPath });
    }
    if (state.title) {
        await chrome.action.setTitle({ title: state.title });
    }
}

/**
 * Animate badge text for "detecting" state
 */
let detectingInterval;
export function startDetectingAnimation() {
    stopDetectingAnimation();
    setIconState('DETECTING');

```

```

    let dots = 0;
    detectingInterval = setInterval(() => {
      dots = (dots + 1) % 4;
      chrome.action.setBadgeText({ text: '.'.repeat(dots) || ' ' });
    }, 400);
  }

  export function stopDetectingAnimation() {
    if (detectingInterval) {
      clearInterval(detectingInterval);
      detectingInterval = null;
    }
  }
}

/**
 * Handle incoming state update requests
 */
chrome.runtime.onMessage.addListener((message, sender,
  sendResponse) => {
  if (message.type === 'UPDATE_BADGE') {
    updateBadgeCount(message.count);
    return false;
  }
  if (message.type === 'SET_ICON_STATE') {
    setIconState(message.state);
    return false;
  }
  if (message.type === 'START_DETECTING') {
    startDetectingAnimation();
    return false;
  }
  if (message.type === 'STOP_DETECTING') {
    stopDetectingAnimation();
    return false;
  }
});

// Optional: Dynamic icon generation via canvas for count-based
// badges
// This is more complex but produces better visuals:
export async function generateBadgeIcon(count) {
  const canvas = new OffscreenCanvas(128, 128);
  const ctx = canvas.getContext('2d');

  // Draw base icon (would load from image in practice)
  ctx.fillStyle = '#1a73e8';
  ctx.beginPath();
  ctx.arc(64, 64, 60, 0, Math.PI * 2);
  ctx.fill();

  // Draw count circle
  if (count > 0) {

```

```

    ctx.fillStyle = '#ef4444';
    ctx.beginPath();
    ctx.arc(96, 32, 28, 0, Math.PI * 2);
    ctx.fill();

    ctx.fillStyle = '#ffffff';
    ctx.font = 'bold 28px sans-serif';
    ctx.textAlign = 'center';
    ctx.textBaseline = 'middle';
    ctx.fillText(String(count > 99 ? 99 : count), 96, 32);
  }

  const imageData = ctx.getImageData(0, 0, 128, 128);
  await chrome.action.setIcon({ imageData });
}

```

## 4.2 Manifest Configuration

```

{
  "action": {
    "default_icon": {
      "16": "icons/icon-16.png",
      "32": "icons/icon-32.png",
      "48": "icons/icon-48.png",
      "128": "icons/icon-128.png"
    },
    "default_title": "Boba Torrent Helper",
    "default_popup": "popup/popup.html"
  }
}

```

## 4.3 API Reference

Claim: `chrome.action.setBadgeText` accepts a string (max ~4 chars visible). `setBadgeBackgroundColor` accepts RGBA array [0-255] or CSS color string. `setIcon` accepts either a path object or `ImageData`.

Source: Chrome Developer Docs

URL: <https://developer.chrome.com/docs/extensions/reference/api/action>

Date: 2025-08-11

Excerpt: "`chrome.action.setBadgeBackgroundColor({color: [0, 255, 0, 0]})... chrome.action.setBadgeText({text: '12'})... The text property is now required in the setBadge method"`

Context: MV3 requires text property; empty string '' clears the badge

Confidence: high

Claim: Badge text max ~4 characters displayed. Badge color can be set per-tab with `tabId` option (auto-resets on tab close).

Source: MDN Web Docs / Chrome Developer Docs

URL: <https://developer.mozilla.org/en-US/docs/Mozilla/Add-ons/WebExtensions/API/action/setBadgeBackgroundColor>  
Date: 2025-11-06  
Excerpt: "Tabs without a specific badge background color will inherit the global badge background color... If a tabId is specified, it removes the tab-specific badge"  
Context: Tab-specific badges useful for showing per-page torrent counts  
Confidence: high

---

## 5. Notifications

### 5.1 Notification Templates

```
// background/notifications.js - Service Worker

/**
 * Notification templates for download lifecycle events
 * Types: basic, image, list, progress
 */

const NOTIFICATION_ICONS = {
  default: chrome.runtime.getURL('icons/icon-128.png'),
  success: chrome.runtime.getURL('icons/icon-success-128.png'),
  error: chrome.runtime.getURL('icons/icon-error-128.png')
};

/**
 * Show "Download Started" notification
 */
export function notifyDownloadStarted(torrentName, torrentId) {
  chrome.notifications.create(`started-${torrentId}`, {
    type: 'basic',
    imageUrl: NOTIFICATION_ICONS.default,
    title: chrome.i18n.getMessage('downloadStarted') || 'Download
      Started',
    message: truncate(torrentName, 80),
    buttons: [
      { title: chrome.i18n.getMessage('viewDashboard') || 'View
        Dashboard' }
    ],
    priority: 0
  });
}

/**
 * Show "Download Complete" notification
 */
export function notifyDownloadComplete(torrentName, torrentId) {
```

```

chrome.notifications.create(`complete-${torrentId}`, {
  type: 'basic',
  imageUrl: NOTIFICATION_ICONS.success,
  title: chrome.i18n.getMessage('downloadComplete') ||
    'Download Complete',
  message: truncate(torrentName, 80),
  buttons: [
    { title: chrome.i18n.getMessage('openFile') || 'Open
      File' },
    { title: chrome.i18n.getMessage('showFolder') || 'Show
      Folder' }
  ],
  priority: 1
});
}

/**
 * Show "Download Error" notification
 */
export function notifyDownloadError(torrentName,
  torrentId, error) {
  chrome.notifications.create(`error-${torrentId}`, {
    type: 'basic',
    imageUrl: NOTIFICATION_ICONS.error,
    title: chrome.i18n.getMessage('downloadError') || 'Download
      Error',
    message: `${truncate(torrentName, 50)}: ${truncate(error, 60)}
      `,
    priority: 2
  });
}

/**
 * Show "Progress" notification (use sparingly - can be noisy)
 */
export function notifyProgress(torrentName, percent, torrentId) {
  // Only update if percent changed significantly (avoid spam)
  const key = `progress-${torrentId}`;
  chrome.notifications.create(key, {
    type: 'progress',
    imageUrl: NOTIFICATION_ICONS.default,
    title: chrome.i18n.getMessage('downloading') || 'Downloading',
    message: truncate(torrentName, 80),
    progress: Math.round(percent),
    priority: 0
  });
}

/**
 * Show "Batch Send" notification (when sending from popup)
 */
export function notifyBatchSent(count) {
  chrome.notifications.create('batch-sent', {

```

```

        type: 'basic',
        imageUrl: NOTIFICATION_ICONS.success,
        title: chrome.i18n.getMessage('sentTitle') || 'Sent to Boba',
        message: chrome.i18n.getMessage('batchSentMessage',
            [String(count)]) ||
            `${count} torrent(s) sent to Boba`,
        priority: 1
    });
}

/**
 * Show "Batch Results" list notification
 */
export function notifyBatchResults(results) {
    const items = results.slice(0, 5).map(r => ({
        title: r.success ? '✓' : '✗',
        message: truncate(r.name, 40)
    }));

    chrome.notifications.create('batch-results', {
        type: 'list',
        imageUrl: NOTIFICATION_ICONS.default,
        title: chrome.i18n.getMessage('sendResults') || 'Send
            Results',
        message: `${results.filter(r => r.success).length}/${
            results.length} successful`,
        items,
        priority: 1
    });
}

// ===== Event Handlers =====

// Handle notification clicks
chrome.notifications.onClicked.addListener((notificationId) => {
    if (notificationId === 'batch-sent' ||
        notificationId.startsWith('started-')) {
        openDashboard();
    }
    // Close the notification
    chrome.notifications.clear(notificationId);
});

// Handle notification button clicks
chrome.notifications.onButtonClicked.addListener((notificationId,
    buttonIndex) => {
    if (notificationId.startsWith('complete-')) {
        const torrentId = notificationId.replace('complete-', '');
        if (buttonIndex === 0) {
            // Open file
            chrome.runtime.sendMessage({ type: 'OPEN_FILE',
                torrentId });
        } else if (buttonIndex === 1) {

```



```

        // Show folder
        chrome.runtime.sendMessage({ type: 'SHOW_FOLDER',
            torrentId });
    }
}
chrome.notifications.clear(notificationId);
});

// Handle notification close
chrome.notifications.onClosed.addListener((notificationId,
    byUser) => {
    // Clean up any tracking state if needed
});

// ===== Helpers =====
function truncate(str, maxLength) {
    return str.length > maxLength ? str.substring(0, maxLength - 3)
        + '...' : str;
}

function openDashboard() {
    chrome.storage.sync.get('bobaServerUrl', (result) => {
        const url = result.bobaServerUrl || 'http://localhost:8080';
        chrome.tabs.create({ url });
    });
}

```

## 5.2 Notification API Reference

Claim: chrome.notifications supports 4 template types: basic, image, list, progress. Buttons (up to 2) supported via buttons array. Events: onClicked, onButtonClicked, onClosed, onPermissionLevelChanged.

Source: Chrome Developer Docs

URL: <https://developer.chrome.com/docs/extensions/reference/api/notifications>

Date: 2026-03-26

Excerpt:

"chrome.notifications.onButtonClicked.addListener((notificationId, buttonIndex) => { ... })... onClicked fires when non-button area clicked"

Context: Progress type shows a progress bar (0-100). Image type includes imageUrl for preview.

Confidence: high

Claim: Platform differences exist: macOS Chrome 59+ shows native notifications instead of Chrome's custom UI. Images are not shown on macOS. List notifications only show first item on macOS.

Source: Chrome Developer Docs (Rich Notifications)

URL: <https://developer.chrome.com/docs/extensions/mv2/richNotifications>

Date: 2014-06-25

Excerpt: "For Mac OS X users on Chrome 59+, images are not shown... list notifications only display the first item"  
Context: Design notifications with these limitations in mind for cross-platform support  
Confidence: high

---

## 6. Keyboard Shortcuts

### 6.1 Manifest Definition

```
{
  "commands": {
    "send-all-torrents": {
      "suggested_key": {
        "default": "Ctrl+Shift+B",
        "mac": "Command+Shift+B"
      },
      "description": "Send all detected torrents to Boba"
    },
    "scan-page": {
      "suggested_key": {
        "default": "Ctrl+Shift+S",
        "mac": "Command+Shift+S"
      },
      "description": "Scan current page for torrents"
    },
    "open-dashboard": {
      "suggested_key": {
        "default": "Ctrl+Shift+D",
        "mac": "Command+Shift+D"
      },
      "description": "Open Boba dashboard"
    },
    "toggle-side-panel": {
      "suggested_key": {
        "default": "Ctrl+Shift+P",
        "mac": "Command+Shift+P"
      },
      "description": "Open Boba side panel"
    },
    "_execute_action": {
      "suggested_key": {
        "default": "Ctrl+Shift+U",
        "mac": "Command+Shift+U"
      },
      "description": "Open Boba popup"
    }
  }
}
```

```
}  
}
```

## 6.2 Command Handler in Service Worker

*// background/commands.js - Service Worker*

```
chrome.commands.onCommand.addListener(async (command) => {  
  console.log(`[Boba] Command received: ${command}`);  
  
  switch (command) {  
    case 'send-all-torrents':  
      await handleSendAll();  
      break;  
  
    case 'scan-page':  
      await handleScanPage();  
      break;  
  
    case 'open-dashboard':  
      await handleOpenDashboard();  
      break;  
  
    case 'toggle-side-panel':  
      await handleToggleSidePanel();  
      break;  
  }  
});  
  
async function handleSendAll() {  
  try {  
    const [tab] = await chrome.tabs.query({ active: true,  
      currentWindow: true });  
    if (!tab?.id) return;  
  
    // Request torrent data from content script  
    const results = await chrome.scripting.executeScript({  
      target: { tabId: tab.id },  
      func: () => {  
        const magnets =  
          Array.from(document.querySelectorAll('a[href^="magnet:"]')).map(a  
            => a.href);  
        const torrents =  
          Array.from(document.querySelectorAll('a[href$=".torrent"]')).map(a  
            => a.href);  
        return [...magnets, ...torrents];  
      }  
    });  
  
    const urls = results[0]?.result || [];  
    if (urls.length === 0) {  
      chrome.notifications.create({
```

```

        type: 'basic',
        iconUrl: chrome.runtime.getURL('icons/icon-128.png'),
        title: 'Boba',
        message: 'No torrents found on this page'
    });
    return;
}

// Send to server
const settings = await
    chrome.storage.sync.get(['bobaServerUrl', 'apiToken']);
const serverUrl = settings.bobaServerUrl || 'http://
    localhost:8080';

let successCount = 0;
for (const url of urls) {
    try {
        await fetch(`${serverUrl}/api/torrents/add`, {
            method: 'POST',
            headers: {
                'Content-Type': 'application/json',
                ...(settings.apiToken && { 'Authorization': `Bearer $
                    {settings.apiToken}` })
            },
            body: JSON.stringify({ url, type:
                url.startsWith('magnet:') ? 'magnet' : 'torrent' })
        });
        successCount++;
    } catch (e) { /* individual failure, continue */ }
}

chrome.notifications.create({
    type: 'basic',
    iconUrl: chrome.runtime.getURL('icons/icon-
        success-128.png'),
    title: 'Boba - Sent',
    message: `${successCount}/${urls.length} torrents sent`
});

} catch (err) {
    console.error('Send all failed:', err);
}
}

async function handleScanPage() {
    // Open popup (which triggers scan) or send message to active
    tab
    const [tab] = await chrome.tabs.query({ active: true,
        currentWindow: true });
    if (tab?.id) {
        chrome.tabs.sendMessage(tab.id, { type: 'SCAN_REQUEST' });
    }
}
}

```

```

async function handleOpenDashboard() {
  const settings = await chrome.storage.sync.get('bobaServerUrl');
  const url = settings.bobaServerUrl || 'http://localhost:8080';
  chrome.tabs.create({ url });
}

async function handleToggleSidePanel() {
  try {
    const [tab] = await chrome.tabs.query({ active: true,
      currentWindow: true });
    if (tab?.windowId) {
      await chrome.sidePanel.open({ windowId: tab.windowId });
    }
  } catch (err) {
    console.error('Side panel toggle failed:', err);
  }
}

```

## 6.3 API Reference

Claim: Commands API allows up to 4 suggested keys per extension. All combinations must include Ctrl or Alt. Special shortcuts: `_execute_action` (MV3), `_execute_browser_action` (MV2). Commands fire `chrome.commands.onCommand` in service worker.

Source: Chrome Developer Docs / MDN

URL: <https://developer.chrome.com/docs/extensions/reference/api/commands>

Date: 2026-04-02

Excerpt: "`_execute_action` (Manifest V3) reserved for triggering action... These commands do not dispatch `command.onCommand` events like standard commands"

Context: `_execute_action` opens popup and does NOT fire `onCommand` - use standard commands for custom actions

Confidence: high

Claim: Supported keys: A-Z, 0-9, Comma, Period, Home, End, PageUp, PageDown, Space, Insert, Delete, Arrow keys, Media Keys. On Mac, 'Ctrl' maps to 'Command' by default; use 'MacCtrl' for actual Control key.

Source: Chrome Extension Docs

URL: <https://sunnyzhou-1024.github.io/chrome-extension-docs/apps/commands.html>

Date: N/A

Excerpt: "On Mac 'Ctrl' is automatically converted to 'Command'. If you want 'Ctrl' instead, please specify 'MacCtrl' under 'mac'"

Context: Critical for cross-platform shortcuts

Confidence: high

---

## 7. Side Panel (Chrome 114+)

### 7.1 Side Panel Implementation

```
<!-- sidepanel/sidepanel.html -->
<!DOCTYPE html>
<html lang="en" data-theme="auto">
<head>
  <meta charset="UTF-8" />
  <meta name="color-scheme" content="light dark" />
  <link rel="stylesheet" href="sidepanel.css" />
</head>
<body>
  <div class="sidepanel">
    <header class="sp-header">
      <div class="sp-brand">
        
        <span class="sp-title" data-il8n="sidePanelTitle">Boba
          Downloads</span>
      </div>
      <div class="sp-connection" id="sp-connection">
        <span class="sp-dot" id="sp-dot"></span>
        <span class="sp-status-text" id="sp-status-text">--</span>
      </div>
    </header>

    <div class="sp-toolbar">
      <button id="sp-refresh" class="sp-btn-icon" aria-
        label="Refresh" title="Refresh">
        <svg width="14" height="14" viewBox="0 0 16 16"
          fill="none" stroke="currentColor" stroke-width="1.5">
          <path d="M13.5 8A5.5 5.5 0 0 1 3.3 10.5M2.5 8a5.5 5.5 0
            0 1 10.2-2.5M11.5 2.5V5h-2.5M4.5 13.5V11h2.5" stroke-
              linecap="round" stroke-linejoin="round"/>
        </svg>
      </button>
      <span class="sp-toolbar-divider"></span>
      <button id="sp-pause-all" class="sp-btn-icon" aria-
        label="Pause all" title="Pause all">
        <svg width="14" height="14" viewBox="0 0 16 16"
          fill="currentColor">
          <rect x="3" y="2" width="4" height="12" rx="1"/>
          <rect x="9" y="2" width="4" height="12" rx="1"/>
        </svg>
      </button>
      <button id="sp-resume-all" class="sp-btn-icon" aria-
        label="Resume all" title="Resume all">
        <svg width="14" height="14" viewBox="0 0 16 16"
          fill="currentColor">
          <path d="M4 2l10 6-10 6V2z"/>
        </svg>
      </button>
    </div>
  </div>
</body>
</html>
```

```

    </button>
  </div>

  <main class="sp-content" id="sp-content">
    <!-- Empty state -->
    <div id="sp-empty" class="sp-empty">
      <p data-il8n="noActiveDownloads">No active downloads</p>
    </div>

    <!-- Download list -->
    <ul id="sp-download-list" class="sp-list" style="display:
      none;"></ul>
  </main>
</div>
<script src="sidepanel.js" type="module"></script>
</body>
</html>

// sidepanel/sidepanel.js

// Side panel has access to all Chrome APIs like popup
let refreshInterval;

// Notify service worker that side panel is ready
document.addEventListener('DOMContentLoaded', () => {
  chrome.runtime.sendMessage({ type: 'SIDE_PANEL_LOADED' });

  document.getElementById('sp-refresh').addEventListener('click',
    refreshDownloads);
  document.getElementById('sp-pause-
    all').addEventListener('click', () =>
    controlAll('pause'));
  document.getElementById('sp-resume-
    all').addEventListener('click', () =>
    controlAll('resume'));

  // Initial load and periodic refresh
  refreshDownloads();
  refreshInterval = setInterval(refreshDownloads, 3000);
});

window.addEventListener('beforeunload', () => {
  if (refreshInterval) clearInterval(refreshInterval);
});

async function refreshDownloads() {
  try {
    const settings = await
      chrome.storage.sync.get('bobaServerUrl');
    const url = settings.bobaServerUrl || 'http://localhost:8080';

    const response = await fetch(`${url}/api/torrents`, {
      signal: AbortSignal.timeout(5000)
    });
  }
}

```

```

    });
    const torrents = await response.json();

    renderDownloads(torrents);
    updateConnection(true, `${torrents.length} active`);
  } catch (err) {
    updateConnection(false, 'Disconnected');
  }
}

function renderDownloads(torrents) {
  const list = document.getElementById('sp-download-list');
  const empty = document.getElementById('sp-empty');

  if (torrents.length === 0) {
    list.style.display = 'none';
    empty.style.display = 'flex';
    return;
  }

  list.style.display = 'flex';
  empty.style.display = 'none';

  list.innerHTML = torrents.map(t => `
    <li class="sp-item" data-id="${t.id}">
      <div class="sp-item-header">
        <span class="sp-item-name" title="${escapeHtml(t.name)}">${
          escapeHtml(t.name)}</span>
        <span class="sp-item-status sp-status-${t.status}">${
          t.status}</span>
      </div>
      <div class="sp-item-progress">
        <div class="sp-progress-bar">
          <div class="sp-progress-fill" style="width: $
            {t.progress}%></div>
        </div>
        <span class="sp-progress-text">${Math.round(t.progress)}
          %</span>
      </div>
      <div class="sp-item-meta">
        <span>${formatBytes(t.downloaded)} / $
          {formatBytes(t.size)}</span>
        <span>${formatSpeed(t.speed)}</span>
      </div>
    </li>
  `).join('');
}

function updateConnection(connected, text) {
  const dot = document.getElementById('sp-dot');
  const statusText = document.getElementById('sp-status-text');
  dot.className = 'sp-dot' + (connected ? ' connected' : '
    disconnected');

```



```

    statusText.textContent = text;
}

function formatBytes(b) {
    if (b === 0) return '0 B';
    const units = ['B', 'KB', 'MB', 'GB', 'TB'];
    const i = Math.floor(Math.log10(b) / 3);
    return (b / Math.pow(1024, i)).toFixed(1) + ' ' + units[i];
}

function formatSpeed(bps) {
    return formatBytes(bps) + '/s';
}

function escapeHtml(text) {
    const d = document.createElement('div');
    d.textContent = text;
    return d.innerHTML;
}

async function controlAll(action) {
    // Send pause/resume all command to server
    const settings = await chrome.storage.sync.get('bobaServerUrl');
    const url = settings.bobaServerUrl || 'http://localhost:8080';
    await fetch(`${url}/api/torrents/${action}all`, { method:
        'POST' });
    refreshDownloads();
}

```

## 7.2 Manifest & Service Worker Configuration

```

{
    "permissions": ["sidePanel"],
    "side_panel": {
        "default_path": "sidepanel/sidepanel.html"
    },
    "action": {
        "default_title": "Click to open Boba side panel"
    }
}

// background/sidePanel.js - Service Worker

import { initContextMenus } from './contextMenus.js';

// Configure action icon to toggle side panel
chrome.sidePanel.setPanelBehavior({ openPanelOnActionClick:
    true })
    .catch(err => console.error('Failed to set panel behavior:',
        err));

// Optional: Open side panel from context menu

```

```

// (see contextMenus.js for the menu item creation)

// Open side panel programmatically (requires user gesture
  context)
export async function openSidePanel(windowId) {
  try {
    await chrome.sidePanel.open({ windowId });
  } catch (err) {
    console.error('Failed to open side panel:', err);
  }
}

// Site-specific side panel (only show on torrent sites)
const TORRENT_SITE_PATTERNS = [
  'https://*.thepiratebay.*/*',
  'https://*.1337x.*/*',
  'https://*.rarbg.*/*',
  'https://*.nyaa.*/*'
];

chrome.tabs.onUpdated.addListener(async (tabId, info, tab) => {
  // Only proceed if we have a URL (page loaded)
  if (!tab.url) return;

  // Optional: Enable/disable side panel based on site
  // By default, side panel is globally available
});

```

## 7.3 Side Panel API Reference

Claim: Side Panel API available in Chrome 114+, MV3 only. Requires "sidePanel" permission. Panel can be opened by: action click (setPanelBehavior), user gesture (sidePanel.open), context menu. Cannot auto-open programmatically without user interaction.

Source: Chrome Developer Docs

URL: <https://developer.chrome.com/docs/extensions/reference/api/sidePanel>

Date: 2026-02-17

Excerpt:

"chrome.sidePanel.setPanelBehavior({ openPanelOnActionClick: true })... sidePanel.open() may only be called in response to a user gesture"

Context: Side panel persists between tabs if configured as global. Tab-specific panels available.

Confidence: high

Claim: sidePanel.open() requires either tabId or windowId.

sidePanel.setOptions() enables/disables panel per-tab.

sidePanel.close() available in Chrome 141+.

Source: Chrome Developer Docs / W3C WebExtensions Issues

URL: <https://github.com/w3c/webextensions/issues/521>

Date: 2024-01-12  
Excerpt: "The sidePanel can be opened via chrome.sidePanel.open() but the API isn't symmetrical, there is no .close() API"  
Context: Workaround for closing: window.close() from within sidepanel.js, or setOptions({enabled: false}) from background  
Confidence: high

---

## 8. Extension Icon States

### 8.1 Icon Set Requirements

Claim: Chrome extension icons should be provided in multiple sizes: 16x16 (favicon/menu), 32x32 (Windows high-DPI), 48x48 (Chrome Web Store), 128x128 (Chrome Web Store listing). Action toolbar uses 16, 24, 32.  
Source: Chrome Developer Docs  
URL: <https://developer.chrome.com/docs/extensions/reference/api/action>  
Date: 2025-08-11  
Excerpt: "action.default\_icon: { '16': 'icon-16.png', '24': 'icon-24.png', '32': 'icon-32.png' }"  
Context: Provide SVG for scalable icons where possible; Chrome will rasterize appropriately  
Confidence: high

### 8.2 Recommended Icon Set

| Icon                 | Purpose                | Sizes           |
|----------------------|------------------------|-----------------|
| icon-*.png           | Default state          | 16, 32, 48, 128 |
| icon-detecting-*.png | Scanning/loading state | 16, 32          |
| icon-sending-*.png   | Sending to server      | 16, 32          |
| icon-error-*.png     | Connection error       | 16, 32          |
| icon-offline-*.png   | Server disconnected    | 16, 32          |
| icon-success-*.png   | Notification: success  | 128             |

### 8.3 Icon State Manager (Complete)

```
// background/iconManager.js

/**
 * Complete icon state manager with SVG-based dynamic icons
 * No external image files needed - generates icons
 *   programmatically
 */
```

```

// Base Boba icon SVG path (a simple download arrow + circle)
const BOBA_PATH = 'M12 2C6.48 2 2 6.48 2 12s4.48 10 10 10 10-4.48
10-10S17.52 2 12 2zm-1 14.5v-4H8l4-4 4
4h-3v4h-2zm1-13c4.14 0 7.5 3.36 7.5 7.5S16.14 19 12
19s-7.5-3.36-7.5-7.5S7.86 4 12 4z';

function createIconCanvas(size, color, badge) {
  const canvas = new OffscreenCanvas(size, size);
  const ctx = canvas.getContext('2d');

  // Clear
  ctx.clearRect(0, 0, size, size);

  // Main icon circle
  ctx.fillStyle = color;
  ctx.beginPath();
  ctx.arc(size / 2, size / 2, size * 0.42, 0, Math.PI * 2);
  ctx.fill();

  // Inner "B" text or symbol
  ctx.fillStyle = '#ffffff';
  ctx.font = `bold ${size * 0.5}px sans-serif`;
  ctx.textAlign = 'center';
  ctx.textBaseline = 'middle';
  ctx.fillText('B', size / 2, size / 2);

  // Badge circle (top-right)
  if (badge) {
    const badgeRadius = size * 0.22;
    const bx = size - badgeRadius - 1;
    const by = badgeRadius + 1;

    ctx.fillStyle = badge.color;
    ctx.beginPath();
    ctx.arc(bx, by, badgeRadius, 0, Math.PI * 2);
    ctx.fill();

    // Badge text
    ctx.fillStyle = '#ffffff';
    const fontSize = badge.text.length > 1 ? size * 0.2 : size *
    0.26;
    ctx.font = `bold ${fontSize}px sans-serif`;
    ctx.fillText(badge.text, bx, by + 1);
  }

  return ctx.getImageData(0, 0, size, size);
}

export async function setDynamicIcon(state, count) {
  const colors = {
    idle: '#5f6368',
    detecting: '#f59e0b',
  }

```

```

    found: '#1a73e8',
    sending: '#f59e0b',
    error: '#ef4444',
    connected: '#10b981',
    disconnected: '#ef4444'
  };

  const color = colors[state] || colors.idle;

  const badge = count > 0 ? {
    color: state === 'error' ? '#ef4444' : '#1a73e8',
    text: count > 99 ? '99' : String(count)
  } : state === 'error' ? {
    color: '#ef4444',
    text: '!'
  } : null;

  const sizes = [16, 32];
  const imageData = {};

  for (const size of sizes) {
    imageData[size] = createIconCanvas(size, color, badge);
  }

  await chrome.action.setIcon({ imageData });

  // Also update badge text separately for accessibility
  if (count > 0) {
    await chrome.action.setBadgeText({
      text: count > 999 ? '999+' : String(count)
    });
    await chrome.action.setBadgeBackgroundColor({ color });
  } else {
    await chrome.action.setBadgeText({ text: '' });
  }
}

```

---

## 9. Dark/Light Theme Support

### 9.1 Theme Detection Methods

Claim: Chrome extensions support dark mode via CSS media query `prefers-color-scheme` and meta color-scheme tag. Use CSS variables for dynamic theme switching. JavaScript can detect via `matchMedia()`.

Source: [text/plain blog](https://textslashplain.com/2021/10/17/spooky-enhancing-dark-mode-in-chromium/) / Chrome Developer Blog

URL: <https://textslashplain.com/2021/10/17/spooky-enhancing-dark-mode-in-chromium/>

Date: 2022-08-03

Excerpt: "To indicate that a page supports dark mode styling, simply add a color-scheme meta tag: <meta name='color-scheme' content='light dark'>... use prefers-color-scheme media query"  
Context: CSS variable approach updates dynamically when OS theme changes without page reload  
Confidence: high

## 9.2 CSS Implementation

```
/* === Complete Theme-Aware CSS Framework === */

/* Base: Light mode */
:root {
  color-scheme: light dark;

  /* Backgrounds */
  --bg-primary: #ffffff;
  --bg-secondary: #f8f9fa;
  --bg-tertiary: #e8eae5;
  --bg-elevated: #ffffff;

  /* Text */
  --text-primary: #202124;
  --text-secondary: #5f6368;
  --text-tertiary: #80868b;
  --text-on-accent: #ffffff;

  /* Accent */
  --accent: #1a73e8;
  --accent-hover: #1557b0;
  --accent-light: #d2e3fc;
  --accent-alpha: rgba(26, 115, 232, 0.12);

  /* Semantic */
  --success: #10b981;
  --success-bg: #d1fae5;
  --error: #ef4444;
  --error-bg: #fee2e2;
  --warning: #f59e0b;
  --warning-bg: #fef3c7;
  --info: #3b82f6;

  /* UI */
  --border: #dadce0;
  --border-hover: #bdc1c6;
  --shadow-sm: 0 1px 2px rgba(60, 64, 67, 0.08);
  --shadow: 0 1px 3px rgba(60, 64, 67, 0.12);
  --shadow-lg: 0 4px 12px rgba(60, 64, 67, 0.15);

  /* Spacing */
  --radius-sm: 4px;
```

```

    --radius: 8px;
    --radius-lg: 12px;
}

/* Dark mode */
@media (prefers-color-scheme: dark) {
  :root {
    --bg-primary: #202124;
    --bg-secondary: #292a2d;
    --bg-tertiary: #3c4043;
    --bg-elevated: #35363a;

    --text-primary: #e8eaed;
    --text-secondary: #9aa0a6;
    --text-tertiary: #80868b;

    --accent: #8ab4f8;
    --accent-hover: #aecbfa;
    --accent-light: #1a1c1e;
    --accent-alpha: rgba(138, 180, 248, 0.12);

    --success: #34d399;
    --success-bg: #064e3b;
    --error: #f87171;
    --error-bg: #450a0a;
    --warning: #fbbf24;
    --warning-bg: #451a03;

    --border: #3c4043;
    --border-hover: #5f6368;
    --shadow-sm: 0 1px 2px rgba(0, 0, 0, 0.3);
    --shadow: 0 1px 3px rgba(0, 0, 0, 0.4);
    --shadow-lg: 0 4px 12px rgba(0, 0, 0, 0.5);
  }
}

/* Force light/dark via data attribute (user override) */
[data-theme="light"] {
  color-scheme: light;
  /* ... override all dark variables to light values ... */
}

[data-theme="dark"] {
  color-scheme: dark;
  /* ... override all light variables to dark values ... */
}

/* Usage: just reference variables */
body {
  background: var(--bg-primary);
  color: var(--text-primary);
}

```

```
.btn-primary {
  background: var(--accent);
  color: var(--text-on-accent);
}

.btn-primary:hover {
  background: var(--accent-hover);
}
```

## 9.3 JavaScript Theme Detection

```
// utils/theme.js

/**
 * Detect and respond to theme changes
 */

export function isDarkMode() {
  return window.matchMedia('(prefers-color-scheme:
    dark)').matches;
}

export function onThemeChange(callback) {
  const mq = window.matchMedia('(prefers-color-scheme: dark)');
  mq.addEventListener('change', (e) => callback(e.matches));
  return () => mq.removeEventListener('change', callback);
}

// Listen for theme changes and update meta if needed
export function initThemeListener() {
  const mq = window.matchMedia('(prefers-color-scheme: dark)');

  // Set initial
  updateMetaThemeColor(mq.matches);

  // Listen for changes
  mq.addEventListener('change', (e) => {
    updateMetaThemeColor(e.matches);
  });
}

function updateMetaThemeColor(isDark) {
  // Update theme-color meta for mobile Chrome
  let meta = document.querySelector('meta[name="theme-color"]');
  if (!meta) {
    meta = document.createElement('meta');
    meta.name = 'theme-color';
    document.head.appendChild(meta);
  }
  meta.content = isDark ? '#202124' : '#ffffff';
}
```



```

}

// For user manual override
export function setTheme(theme) {
  document.documentElement.setAttribute('data-theme', theme);
  chrome.storage.sync.set({ theme });
}

export async function loadThemePreference() {
  const { theme } = await chrome.storage.sync.get('theme');
  if (theme === 'light' || theme === 'dark') {
    document.documentElement.setAttribute('data-theme', theme);
  }
  // 'auto' or undefined = follow system
}

```

---

## 10. Responsive Design

### 10.1 Popup Constraints

Claim: Chrome popup max size: 800x600px (Chromium source: gfx::Size kMaxSize = {800, 600}). Recommended popup width: 380-450px. Height should be content-driven with max-height and overflow-y: auto for scrolling.

Source: Extension Booster / Chromium Source

URL: <https://extensionbooster.net/blog/chrome-extension-popup-ui-design-best-practices-guide/>

Date: 2026-04-23

Excerpt: "The sweet spot that top extensions actually use is 400x500px. It fits comfortably in peripheral vision without covering the page content underneath."

Context: Define explicit width on both html and body elements. Never rely on width: 100%.

Confidence: high

### 10.2 Responsive CSS Patterns

```

/* popup/popup.css - Responsive additions */

/* Base popup sizing */
html, body {
  width: 400px;
  min-height: 300px;
  max-height: 580px;
  overflow: hidden;
}

/* Mobile-friendly touch targets */

```

```

button, .torrent-item, select, input[type="checkbox"] {
  min-height: 32px;
  min-width: 32px;
}

/* Scrollbar styling for consistency */
.popup-content::-webkit-scrollbar {
  width: 6px;
}
.popup-content::-webkit-scrollbar-thumb {
  background: var(--border-color);
  border-radius: 3px;
}
.popup-content::-webkit-scrollbar-thumb:hover {
  background: var(--text-tertiary);
}

/* Handle long text with ellipsis */
.torrent-name {
  white-space: nowrap;
  overflow: hidden;
  text-overflow: ellipsis;
  max-width: 280px;
}

/* Flexbox-based responsive layout */
.torrent-list-header {
  display: flex;
  flex-wrap: wrap;
  gap: 8px;
  align-items: center;
  justify-content: space-between;
}

/* Ensure footer doesn't get pushed off-screen */
#app {
  display: flex;
  flex-direction: column;
  height: 100vh;
  max-height: 580px;
}

.popup-content {
  flex: 1;
  min-height: 0; /* Critical: allows flex child to shrink below
                  content size */
  overflow-y: auto;
}

/* Density modes */
[data-density="compact"] .torrent-item {
  padding: 6px 10px;
}

```

```

}
[data-density="comfortable"] .torrent-item {
  padding: 14px 16px;
}

/* For high-DPI displays */
@media (-webkit-min-device-pixel-ratio: 2) {
  .header-icon,
  .torrent-checkbox {
    image-rendering: -webkit-optimize-contrast;
  }
}

```

---

## 11. Accessibility

### 11.1 ARIA Labels & Roles

Claim: All interactive elements must have accessible names. Use `aria-label` for icon-only buttons, `aria-labelledby` for grouped controls, `role` for custom widgets. Focus indicators must be visible.

Source: AllAccessible / Chrome Extension UI Guide

URL: <https://www.allaccessible.org/blog/implementing-aria-labels-for-web-accessibility>

Date: 2024-01-18

Excerpt: "aria-label: Provides accessible name for elements...  
`role='dialog'`: Modal dialog... `aria-live='polite'`: Announce updates when convenient"

Context: Screen reader users need context for all interactive elements

Confidence: high

### 11.2 Accessibility Implementation

*<!-- Key accessibility patterns -->*

*<!-- Icon-only buttons MUST have aria-label -->*

```

<button class="icon-btn" aria-label="Send to Boba" title="Send to Boba">

```

```

  <svg aria-hidden="true">...</svg>

```

```

</button>

```

*<!-- Live regions for dynamic content -->*

```

<div id="status" role="status" aria-live="polite" class="sr-only"></div>

```

*<!-- Listbox pattern for torrent list -->*

```

<ul role="listbox" aria-label="Detected torrents" aria-multiselectable="true">

```

```

    <li role="option" aria-selected="false" tabindex="0">...
</ul>

<!-- Progress indicators with ARIA -->
<div role="progressbar"
    aria-valuemin="0"
    aria-valuemax="100"
    aria-valuenow="42"
    aria-label="Upload progress">
    <div class="progress-bar"><div class="progress-fill"
        style="width: 42%"></div></div>
</div>

<!-- Ensure form labels are associated -->
<label for="server-url">Server URL</label>
<input id="server-url" type="url" aria-describedby="url-help" />
<p id="url-help">Enter the Boba server URL</p>

/* Accessibility additions */

/* Visually hidden (screen reader only) */
.sr-only {
    position: absolute;
    width: 1px;
    height: 1px;
    padding: 0;
    margin: -1px;
    overflow: hidden;
    clip: rect(0, 0, 0, 0);
    white-space: nowrap;
    border: 0;
}

/* Visible focus indicators */
:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 2px;
}

/* Skip link for keyboard navigation */
.skip-link {
    position: absolute;
    top: -40px;
    left: 0;
    background: var(--accent);
    color: #fff;
    padding: 8px;
    text-decoration: none;
    z-index: 100;
}
.skip-link:focus {
    top: 0;
}

```

```

}

/* Ensure sufficient color contrast (WCAG AA: 4.5:1 for text) */
/* Our variable choices above meet this standard */

/* Reduced motion preference */
@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after {
    animation-duration: 0.01ms !important;
    animation-iteration-count: 1 !important;
    transition-duration: 0.01ms !important;
  }
}

/* High contrast mode */
@media (prefers-contrast: high) {
  .torrent-item {
    border-width: 2px;
  }
  .btn {
    border: 2px solid currentColor;
  }
}

// utils/ally.js - Accessibility utilities

/**
 * Announce a message to screen readers
 */
export function announce(message, priority = 'polite') {
  let announcer = document.getElementById('ally-announcer');
  if (!announcer) {
    announcer = document.createElement('div');
    announcer.id = 'ally-announcer';
    announcer.className = 'sr-only';
    announcer.setAttribute('aria-live', priority);
    announcer.setAttribute('aria-atomic', 'true');
    document.body.appendChild(announcer);
  }

  // Clear and set (forces re-announcement)
  announcer.textContent = '';
  requestAnimationFrame(() => {
    announcer.textContent = message;
  });
}

/**
 * Trap focus within an element (modal dialog pattern)
 */
export function trapFocus(element) {
  const focusable = element.querySelectorAll(

```

```

        'button, [href], input, select, textarea,
          [tabindex]:not([tabindex="-1"])'
    );
    const first = focusable[0];
    const last = focusable[focusable.length - 1];

    element.addEventListener('keydown', (e) => {
        if (e.key !== 'Tab') return;

        if (e.shiftKey && document.activeElement === first) {
            e.preventDefault();
            last.focus();
        } else if (!e.shiftKey && document.activeElement === last) {
            e.preventDefault();
            first.focus();
        }
    });
}

/**
 * Keyboard navigation for custom list
 */
export function setupListKeyboardNavigation(listElement, options
    = {}) {
    const items = () =>
        listElement.querySelectorAll('[role="option"]');

    listElement.addEventListener('keydown', (e) => {
        const current = document.activeElement;
        const itemList = Array.from(items());
        const index = itemList.indexOf(current);

        switch (e.key) {
            case 'ArrowDown':
                e.preventDefault();
                const next = itemList[index + 1] || itemList[0];
                next?.focus();
                break;
            case 'ArrowUp':
                e.preventDefault();
                const prev = itemList[index - 1] ||
                    itemList[itemList.length - 1];
                prev?.focus();
                break;
            case 'Home':
                e.preventDefault();
                itemList[0]?.focus();
                break;
            case 'End':
                e.preventDefault();
                itemList[itemList.length - 1]?.focus();
                break;
        }
    });
}

```

```

        case ' ':
            e.preventDefault();
            // Toggle selection
            current.click();
            break;
    }
});
}

```

---

## 12. Internationalization (i18n)

### 12.1 Directory Structure

```

extension-root/
  _locales/
    en/
      messages.json
    es/
      messages.json
    fr/
      messages.json
    de/
      messages.json
    zh_CN/
      messages.json
  manifest.json
  ...

```

### 12.2 messages.json (English - Complete Reference)

```

{
  "extName": {
    "message": "Boba Torrent Helper",
    "description": "Extension display name"
  },
  "extDescription": {
    "message":
      "Detect and send torrents/magnet links to your Boba
      server",
    "description": "Extension description in store"
  },
  "popupTitle": {
    "message": "Boba",
    "description": "Popup header title"
  },
  "noTorrentsFound": {
    "message": "No torrents found",

```

```

    "description": "Empty state title"
  },
  "scanThisPage": {
    "message": "Click refresh to scan this page",
    "description": "Empty state subtitle"
  },
  "scanError": {
    "message": "Scan error",
    "description": "Error state title"
  },
  "tryAgain": {
    "message": "Click to try again",
    "description": "Error state subtitle"
  },
  "torrentsFound": {
    "message": "$count$ torrents found",
    "description": "Torrent count label",
    "placeholders": {
      "count": {
        "content": "$1",
        "example": "5"
      }
    }
  },
  "sendToBoba": {
    "message": "Send to Boba",
    "description": "Send button label"
  },
  "sendAllToBoba": {
    "message": "Send All to Boba",
    "description": "Send all button label"
  },
  "sendingTorrents": {
    "message": "Sending to Boba...",
    "description": "Sending state title"
  },
  "sendingItem": {
    "message": "$current$ of $total$: $name$",
    "description": "Current sending progress detail",
    "placeholders": {
      "current": { "content": "$1", "example": "1" },
      "total": { "content": "$2", "example": "5" },
      "name": { "content": "$3", "example": "filename.torrent" }
    }
  },
  "sentSuccessfully": {
    "message": "Sent successfully!",
    "description": "Success state title"
  },
  "sentDetail": {
    "message": "$count$ torrent(s) sent",
    "description": "Success detail text",

```



```
    "placeholders": {
      "count": { "content": "$1", "example": "3" }
    },
    "viewInDashboard": {
      "message": "View in Dashboard",
      "description": "View dashboard button"
    },
    "openDashboard": {
      "message": "Dashboard",
      "description": "Dashboard link"
    },
    "connecting": {
      "message": "Connecting...",
      "description": "Connection status"
    },
    "connected": {
      "message": "Connected",
      "description": "Connected status"
    },
    "notConnected": {
      "message": "Not connected",
      "description": "Disconnected status"
    },
    "serverError": {
      "message": "Server error",
      "description": "Server error status"
    },
    "scanning": {
      "message": "Scanning...",
      "description": "Scanning status"
    },
    "downloadStarted": {
      "message": "Download Started",
      "description": "Notification title"
    },
    "downloadComplete": {
      "message": "Download Complete",
      "description": "Notification title"
    },
    "downloadError": {
      "message": "Download Error",
      "description": "Notification title"
    },
    "downloading": {
      "message": "Downloading",
      "description": "Progress notification title"
    },
    "sentTitle": {
      "message": "Sent to Boba",
```

```

    "description": "Notification title"
  },
  "sendMessage": {
    "message": "$count$ torrent(s) sent to Boba",
    "description": "Notification message",
    "placeholders": {
      "count": { "content": "$1", "example": "3" }
    }
  },
  "sendFailed": {
    "message": "Failed to send",
    "description": "Error notification title"
  },
  "linkSent": {
    "message": "Link sent to Boba",
    "description": "Link sent confirmation"
  },
  "viewDashboard": {
    "message": "View Dashboard",
    "description": "Notification button"
  },
  "openFile": {
    "message": "Open File",
    "description": "Notification button"
  },
  "showFolder": {
    "message": "Show Folder",
    "description": "Notification button"
  },
  "optionsTitle": {
    "message": "Boba Settings",
    "description": "Options page title"
  },
  "serverConnection": {
    "message": "Server Connection",
    "description": "Settings section title"
  },
  "serverConnectionDesc": {
    "message": "Configure your Boba server URL and API
      credentials.",
    "description": "Settings section description"
  },
  "serverUrlLabel": {
    "message": "Boba Server URL",
    "description": "Settings label"
  },
  "serverUrlHelp": {
    "message":
      "The URL where your Boba server is running. Include the
      port if not using standard ports.",
    "description": "Settings help text"
  }

```

```
,
"apiTokenLabel": {
  "message": "API Token (Optional)",
  "description": "Settings label"
},
"apiTokenHelp": {
  "message":
    "If your Boba server requires authentication, enter your
    API token here.",
  "description": "Settings help text"
},
"testConnection": {
  "message": "Test Connection",
  "description": "Button label"
},
"downloadSettings": {
  "message": "Download Settings",
  "description": "Settings section title"
},
"defaultCategory": {
  "message": "Default Category",
  "description": "Settings label"
},
"categoryNone": {
  "message": "None",
  "description": "Category option"
},
"categoryMovies": {
  "message": "Movies",
  "description": "Category option"
},
"categoryTV": {
  "message": "TV Shows",
  "description": "Category option"
},
"categoryMusic": {
  "message": "Music",
  "description": "Category option"
},
"categorySoftware": {
  "message": "Software",
  "description": "Category option"
},
"categoryOther": {
  "message": "Other",
  "description": "Category option"
},
"autoStart": {
  "message": "Auto-start downloads",
  "description": "Settings label"
},
"autoStartHelp": {
```

```
    "message":
      "Automatically start downloading torrents when sent to
      Boba.",
    "description": "Settings help text"
  },
  "enableNotifications": {
    "message": "Enable notifications",
    "description": "Settings label"
  },
  "notificationsHelp": {
    "message": "Show browser notifications for download events.",
    "description": "Settings help text"
  },
  "behavior": {
    "message": "Behavior",
    "description": "Settings section title"
  },
  "scanMode": {
    "message": "Scan Mode",
    "description": "Settings label"
  },
  "scanManual": {
    "message": "Manual (click to scan)",
    "description": "Scan mode option"
  },
  "scanAuto": {
    "message": "Automatic (on page load)",
    "description": "Scan mode option"
  },
  "contextMenu": {
    "message": "Show context menu items",
    "description": "Settings label"
  },
  "dangerZone": {
    "message": "Danger Zone",
    "description": "Settings section title"
  },
  "resetSettings": {
    "message": "Reset All Settings",
    "description": "Button label"
  },
  "resetConfirm": {
    "message": "Reset all settings to defaults?",
    "description": "Confirmation dialog"
  },
  "saveSettings": {
    "message": "Save Settings",
    "description": "Button label"
  },
  "saved": {
    "message": "Saved!",
    "description": "Save status"
  }
```

```

},
"saveFailed": {
  "message": "Save failed",
  "description": "Save error status"
},

"testing": {
  "message": "Testing...",
  "description": "Connection test status"
},
"connectionSuccess": {
  "message": "Connected!",
  "description": "Connection test result"
},
"connectionFailed": {
  "message": "Server error",
  "description": "Connection test result"
},
"connectionError": {
  "message": "Cannot connect",
  "description": "Connection test result"
},

"sidePanelTitle": {
  "message": "Boba Downloads",
  "description": "Side panel title"
},
"noActiveDownloads": {
  "message": "No active downloads",
  "description": "Empty state text"
}
}

```

## 12.3 Manifest Configuration

```

{
  "default_locale": "en",
  "name": "__MSG_extName__",
  "description": "__MSG_extDescription__"
}

```

## 12.4 JavaScript i18n Usage

```

// In any extension page (popup, options, sidepanel):

// Get a localized string
const title = chrome.i18n.getMessage('popupTitle');

// With substitutions
const count = 5;

```

```

const message = chrome.i18n.getMessage('torrentsFound',
    [String(count)]);
// Result: "5 torrents found"

// Helper function for templates
function t(key, ...substitutions) {
    return chrome.i18n.getMessage(key, substitutions.map(String))
        || key;
}

// Apply to all data-i18n elements
document.querySelectorAll('[data-i18n]').forEach(el => {
    const key = el.dataset.i18n;
    const msg = chrome.i18n.getMessage(key);
    if (msg) el.textContent = msg;
});

// Get user's Accept-Languages
chrome.i18n.getAcceptLanguages((languages) => {
    console.log('User languages:', languages); // ['en-US', 'en',
        'es']
});

// Get current UI language
console.log(chrome.i18n.getUILanguage()); // 'en-US'

```

## 12.5 API Reference

Claim: i18n requires `_locales` directory with `messages.json` files. manifest MUST define "default\_locale" if `_locales` exists. Strings referenced via `__MSG_key__` in manifest/CSS, `chrome.i18n.getMessage()` in JS. Supports placeholders up to 9 substitutions.

Source: Chrome Developer Docs

URL: <https://developer.chrome.com/docs/extensions/reference/api/i18n>

Date: 2026-05-09

Excerpt: "If an extension has a `_locales` directory, the manifest must define 'default\_locale'... In JavaScript:

`chrome.i18n.getMessage('messagename')`"

Context: Fallback chain: user locale -> base locale (e.g., `en_GB` -> `en`) -> `default_locale`

Confidence: high

Claim: Predefined messages available: `@@ui_locale`, `@@bidi_dir`, `@@bidi_reversed_dir`, `@@bidi_start_edge`, `@@bidi_end_edge`. Useful for RTL support.

Source: Chromium i18n Documentation

URL: <https://www.chromium.org/developers/design-documents/extensions/how-the-extension-system-works/i18n/>

Date: N/A

Excerpt: "`chrome.i18n.getMessage('@@bidi_dir')` returns `'ltr'` or

'rtl''

Context: Use for RTL layout adjustments

Confidence: high

---

## Appendix A: Complete manifest.json

```
{
  "manifest_version": 3,
  "name": "__MSG_extName__",
  "description": "__MSG_extDescription__",
  "version": "1.0.0",
  "default_locale": "en",

  "permissions": [
    "storage",
    "activeTab",
    "scripting",
    "contextMenus",
    "notifications",
    "sidePanel"
  ],

  "host_permissions": [
    "http://localhost:*/",
    "https://*/"
  ],

  "action": {
    "default_icon": {
      "16": "icons/icon-16.png",
      "32": "icons/icon-32.png",
      "48": "icons/icon-48.png",
      "128": "icons/icon-128.png"
    },
    "default_title": "Boba Torrent Helper",
    "default_popup": "popup/popup.html"
  },

  "background": {
    "service_worker": "background/service-worker.js",
    "type": "module"
  },

  "options_page": "options/options.html",

  "side_panel": {
    "default_path": "sidepanel/sidepanel.html"
  },
}
```

```
"commands": {
  "send-all-torrents": {
    "suggested_key": {
      "default": "Ctrl+Shift+B",
      "mac": "Command+Shift+B"
    },
    "description": "Send all detected torrents to Boba"
  },
  "scan-page": {
    "suggested_key": {
      "default": "Ctrl+Shift+S",
      "mac": "Command+Shift+S"
    },
    "description": "Scan current page for torrents"
  },
  "open-dashboard": {
    "suggested_key": {
      "default": "Ctrl+Shift+D",
      "mac": "Command+Shift+D"
    },
    "description": "Open Boba dashboard"
  },
  "toggle-side-panel": {
    "suggested_key": {
      "default": "Ctrl+Shift+P",
      "mac": "Command+Shift+P"
    },
    "description": "Toggle Boba side panel"
  }
},

"icons": {
  "16": "icons/icon-16.png",
  "32": "icons/icon-32.png",
  "48": "icons/icon-48.png",
  "128": "icons/icon-128.png"
},

"web_accessible_resources": [
  {
    "resources": ["icons/*"],
    "matches": ["<all_urls>"]
  }
]
}
```

---



## Appendix B: File Structure

```
boba-extension/  
  manifest.json  
  
  _locales/  
    en/  
      messages.json  
  
  background/  
    service-worker.js      # Main service worker entry  
    contextMenus.js        # Context menu registration  
    iconState.js           # Badge/icon state management  
    notifications.js       # Notification handlers  
    commands.js           # Keyboard shortcut handlers  
    sidePanel.js           # Side panel configuration  
  
  popup/  
    popup.html             # Popup HTML  
    popup.css              # Popup styles (theme-aware)  
    popup.js               # Popup logic  
  
  options/  
    options.html           # Settings page HTML  
    options.css            # Settings styles  
    options.js             # Settings logic  
  
  sidepanel/  
    sidepanel.html         # Side panel HTML  
    sidepanel.css          # Side panel styles  
    sidepanel.js           # Side panel logic  
  
  icons/  
    icon-16.png            # Toolbar icon  
    icon-32.png            # Toolbar @2x  
    icon-48.png            # Chrome Web Store  
    icon-128.png           # Store listing  
    icon-success-128.png   # Success notification  
    icon-error-128.png     # Error notification  
  
  utils/  
    theme.js               # Theme detection utilities  
    ally.js                # Accessibility utilities  
    i18n.js                # i18n helper functions
```

---

# Appendix C: Key API Version Reference

| API                      | Min Chrome | MV Version  | Notes                             |
|--------------------------|------------|-------------|-----------------------------------|
| chrome.action            | 88         | MV3         | Replaces browserAction/pageAction |
| chrome.storage.session   | 102        | MV3         | In-memory session storage         |
| chrome.sidePanel         | 114        | MV3         | Side panel for persistent UI      |
| chrome.sidePanel.open()  | 116        | MV3         | Programmatic open                 |
| chrome.sidePanel.close() | 141        | MV3         | Programmatic close                |
| chrome.notifications     | 28         | MV2/<br>MV3 | Rich notifications                |
| chrome.contextMenus      | 6          | MV2/<br>MV3 | Right-click menus                 |
| chrome.commands          | 35         | MV2/<br>MV3 | Keyboard shortcuts                |
| chrome.i18n              | 1          | MV2/<br>MV3 | Internationalization              |
| chrome.storage.sync      | 1          | MV2/<br>MV3 | Cross-device sync (100KB)         |
| chrome.storage.local     | 1          | MV2/<br>MV3 | Local storage (10MB)              |

*Research compiled from 20+ authoritative sources including official Chrome Developer documentation, MDN Web Docs, Chromium source code, and established extension development guides. All code examples are written for Manifest V3 and target Chrome 114+.*